Foundations and Applications for a

CERTIFIED PERSONAL TRAINER WORKBOOK



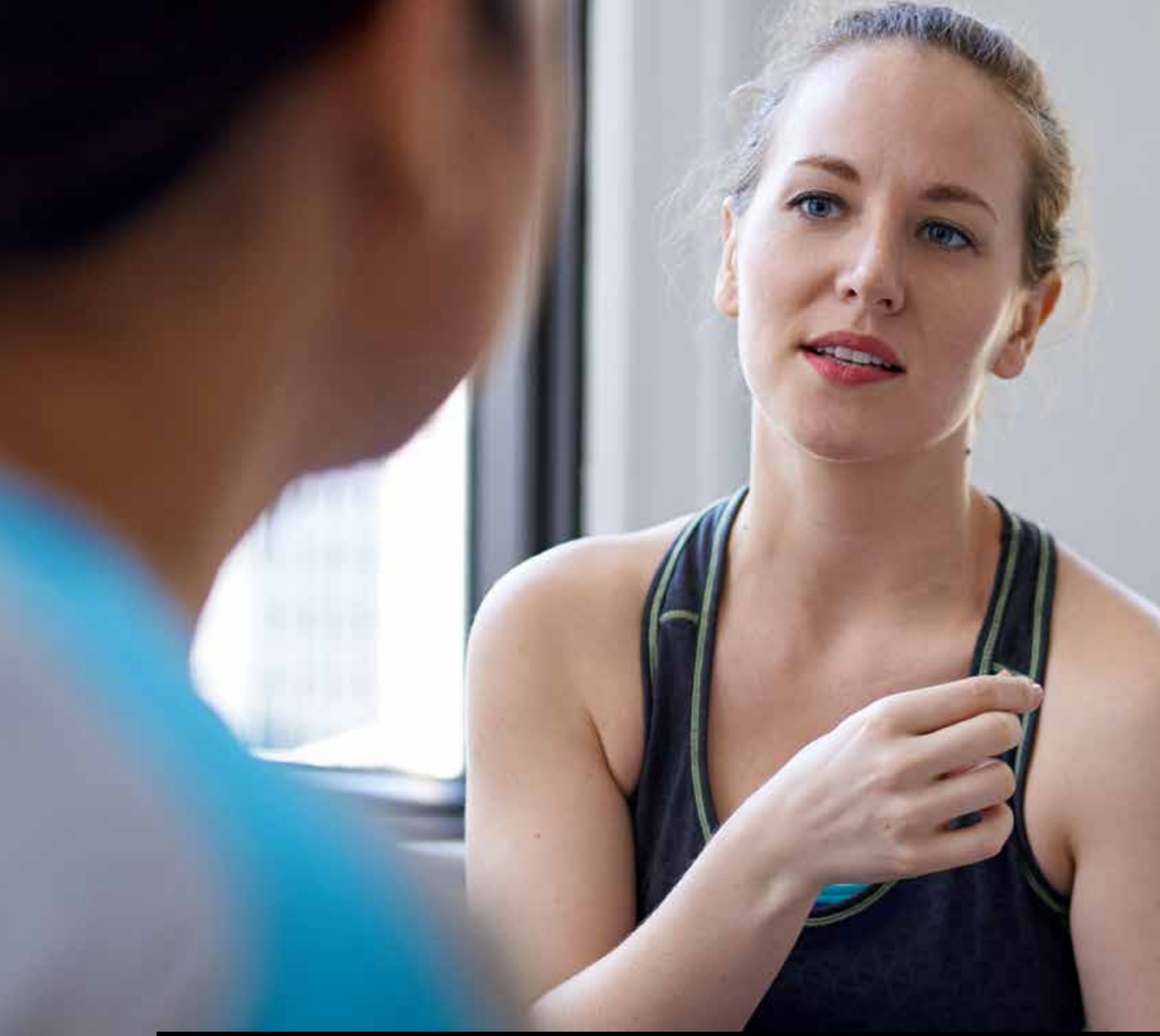
ISSA INTERNATIONAL SPORTS SCIENCES ASSOCIATION

ABOUT THE WORKBOOK

This Workbook serves as a resource companion to your personal training textbook. The Workbook has been designed with one main purpose: to help you better understand the content of the main text. This is accomplished by focusing on important points and concepts, and having you write the answers directly in the guide. This way of studying also provides an easy way to follow the flow of the course text. If you follow along with the Workbook, you will have covered ALL of the main topics of the course text.

Students employ different strategies to learn course materials and to prepare for the CPT Certification Exam. The most commonly used strategy employs reading and underlining text material; another includes outlining or writing key concepts. This Workbook encourages each of these methods.

Experience has shown that individuals who faithfully use the Workbook achieve greater understanding of the course materials than those who do not use a Workbook.



CHAPTER 02

PSYCHOLOGY OF BEHAVIOR CHANGE

Fill in the following table.

STAGE OF CHANGE	DESCRIPTION	KEY INDICATORS
		Lacks the belief they could change (quit a behavior or start a behavior), even if they wanted to Perceives no control over their behavior (e.g., resigned to their current state) Defeated by failed prior attempts to change
		Aware of potential benefits of making a change but perceive the costs (e.g., time, effort, sacrifices) outweighing the benefits Procrastinating making efforts toward the behavior change
		Begins to take small steps toward the target behavior change (e.g., buys exercise apparel or equipment, signs up for a gym membership, collects information, initiates the hire of a personal trainer)
		Actively doing things to change or modify behavior Structures their environment in ways that support their healthy behavior (e.g., avoids temptations that trigger the undesirable behavior)
		New behavior becomes second nature Greater confidence in ability to maintain the new behavior Greater sense of control over their behavior

WORKBOOK

Name and describe the 10 processes of change a personal trainer can use with clients.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____

Define "motivation."

_____

_____

_____

_____

What is the difference between intrinsic motivation and extrinsic motivation?

_____

_____

_____

_____

Define the 3 basic human needs.

PSYCHOLOGICAL NEED	DESCRIPTION
Autonomy	
Competence	
Relatedness	

Explain the OARS Model and what it is used for.

O - _____

A - _____

R - _____

S - _____

Define and describe an "outcome goal" and a "process goal."

_____

_____

_____

In goal setting, what does the acronym SMART stand for?

S - _____

M - _____

A - _____

R - _____

T - _____

Give an example of a SMART goal related to fitness.

_____

_____

_____

_____

Explain the importance of feedback between a personal trainer and client.

_____

_____

_____

_____

_____

_____



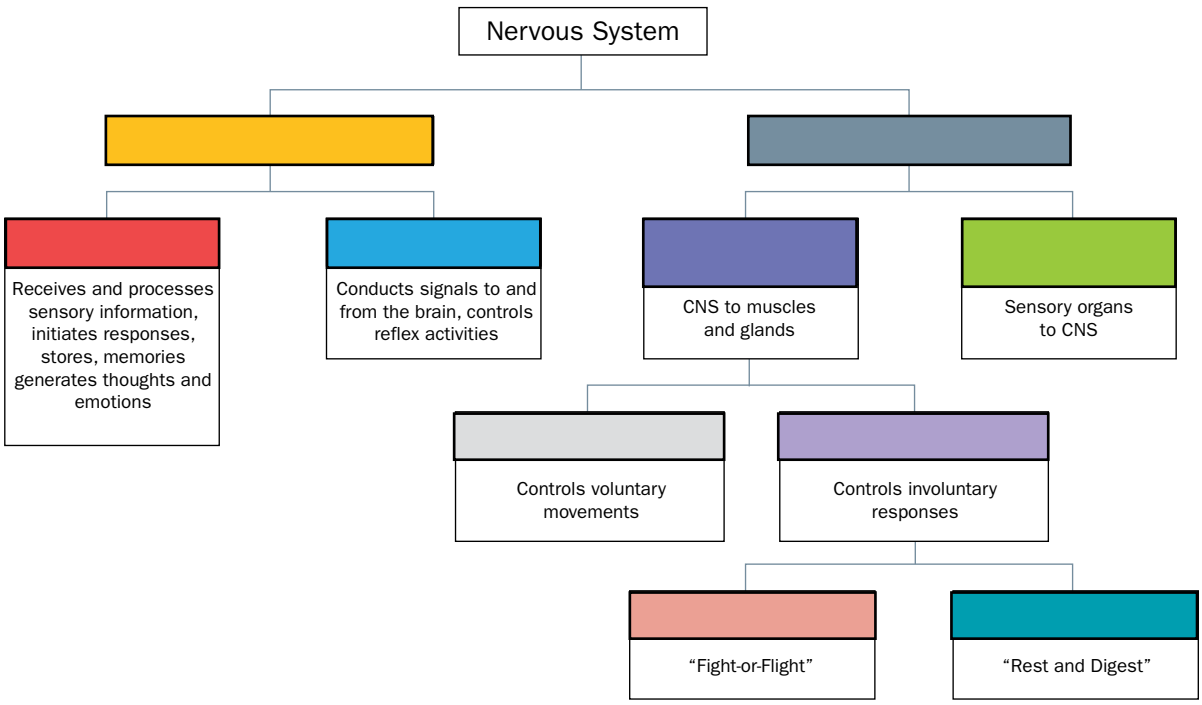
CHAPTER 3

MOVEMENT SYSTEMS

List the levels of organization in the human body.

Explain the nervous system and its two major parts.

Fill in the following flow chart of the nervous system.



Name and explain the three types of muscle tissue found in the human body.

1. _____

2. _____

3. _____

Name and describe the types of muscle fibers.

_____

_____

_____

Fill in the names of the types of muscles.



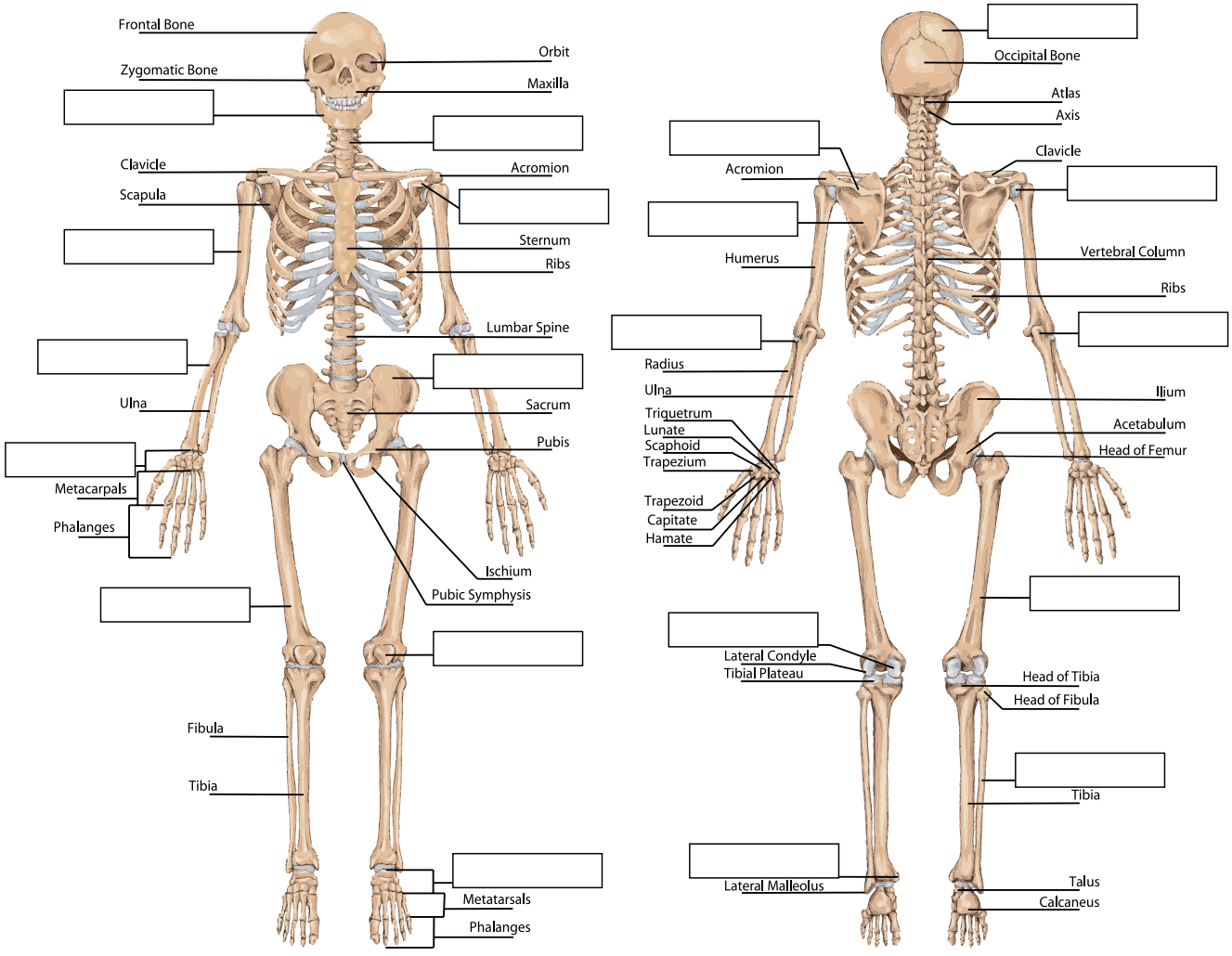
Write the number of bones that compose each of the following:

Overall skeleton: _____

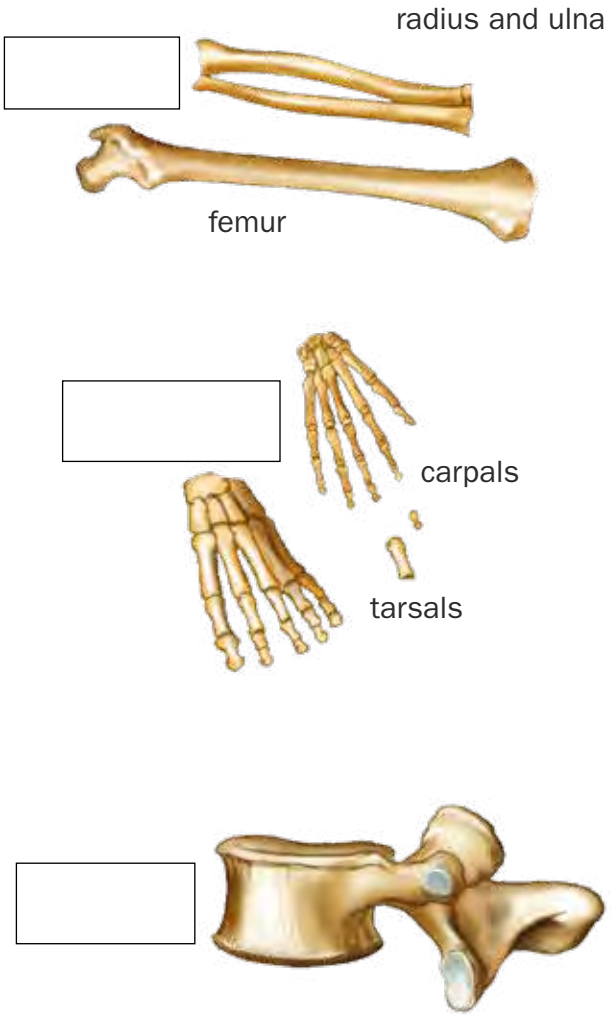
Axial skeleton: _____

Appendicular skeleton: _____

Label the following skeletal chart.

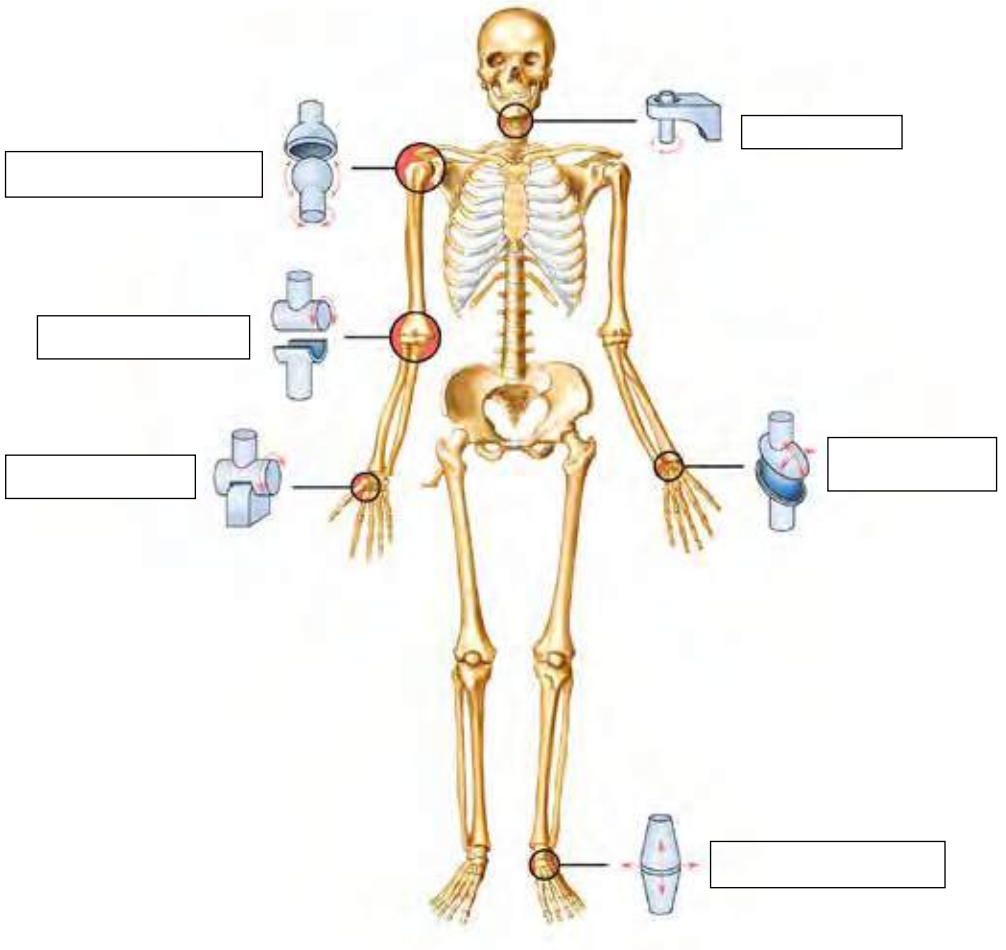


Fill in the classifications of the following bones.





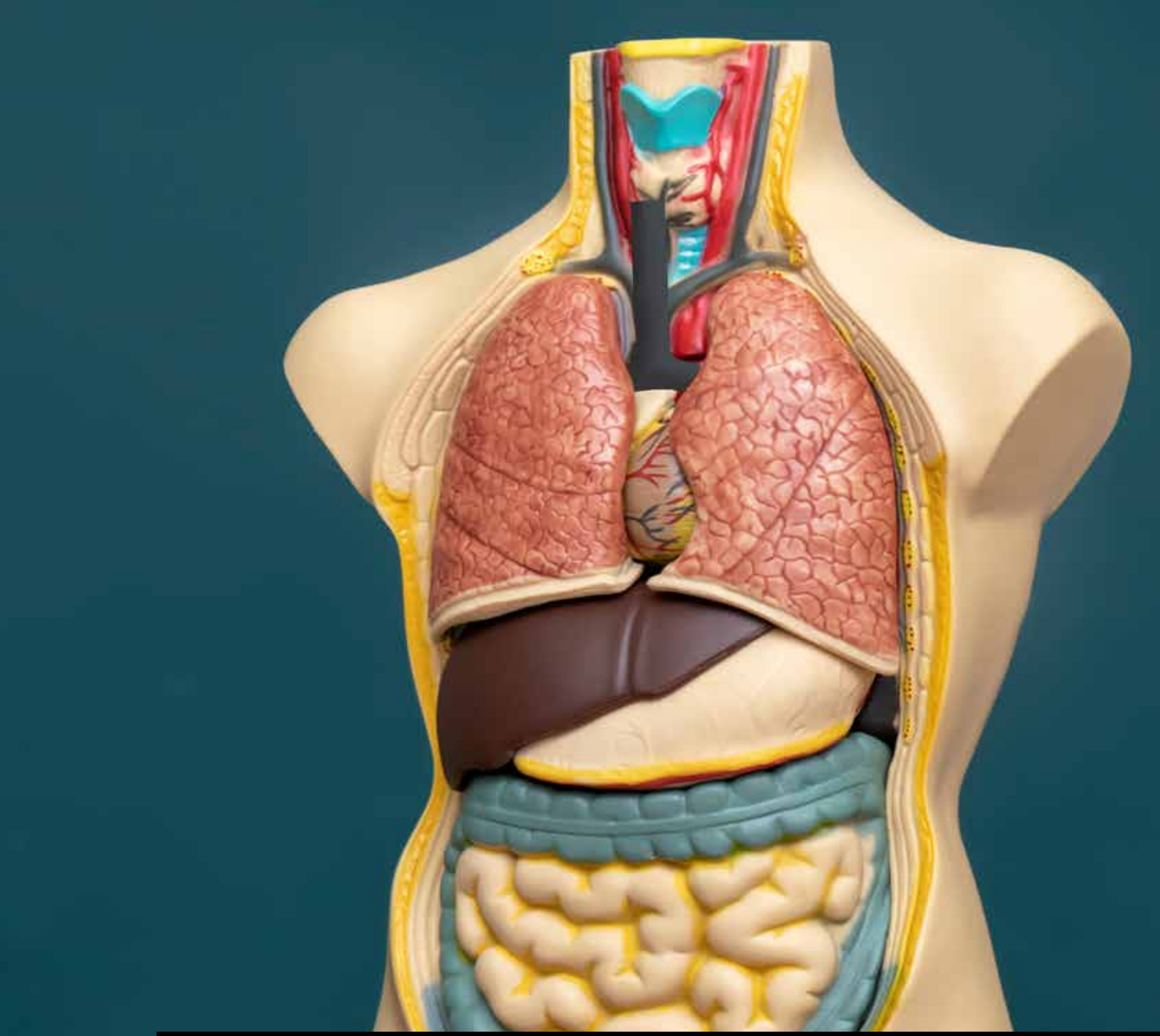
Label the types of joints in the body.



Define "tendon."

Define "joint."

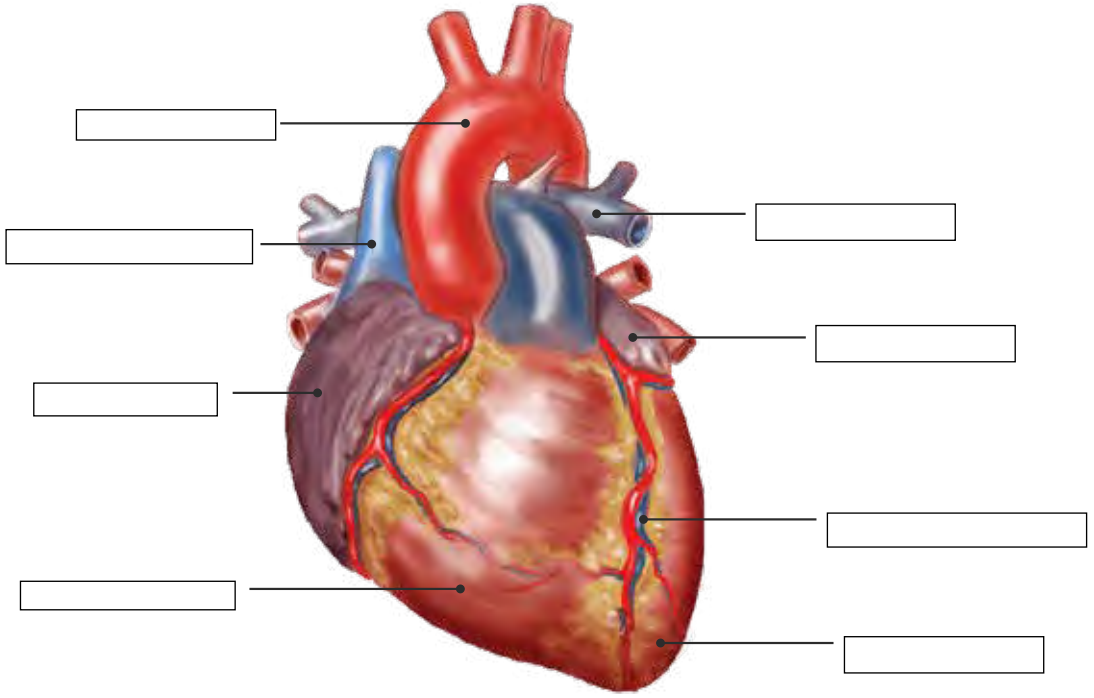
Define "ligament."

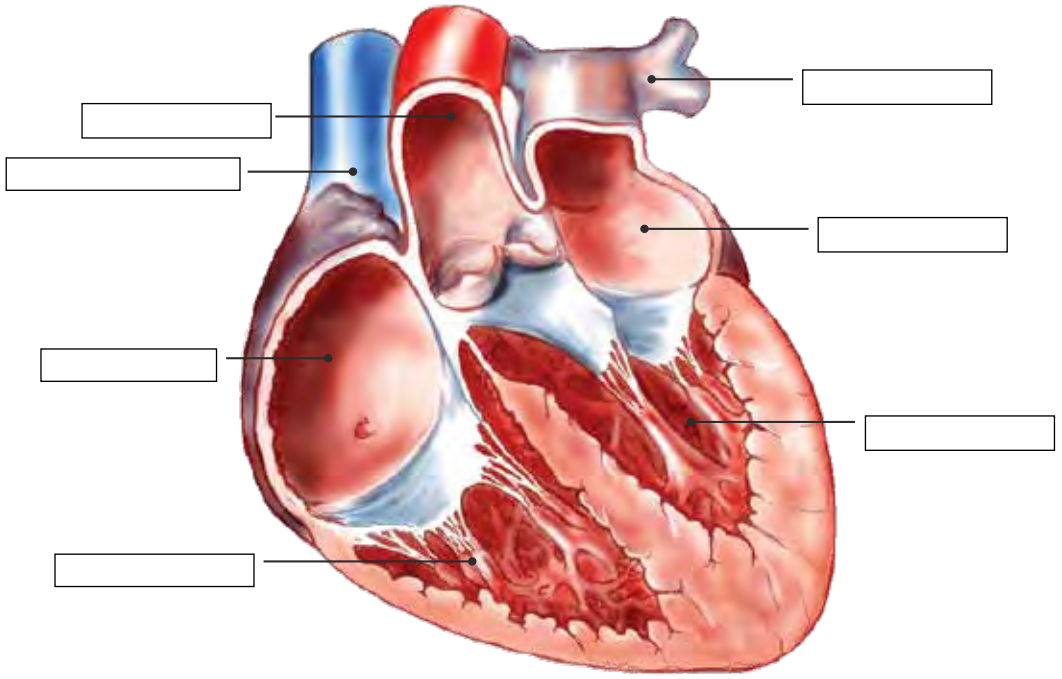


CHAPTER 4

SUPPORTING SYSTEMS

Label the heart diagrams below.





Explain the digestive system and its role in our bodies.

Explain the respiratory system and its role in our bodies.

Explain the circulatory system and its role in our bodies.

Explain the role of hormones in the body.

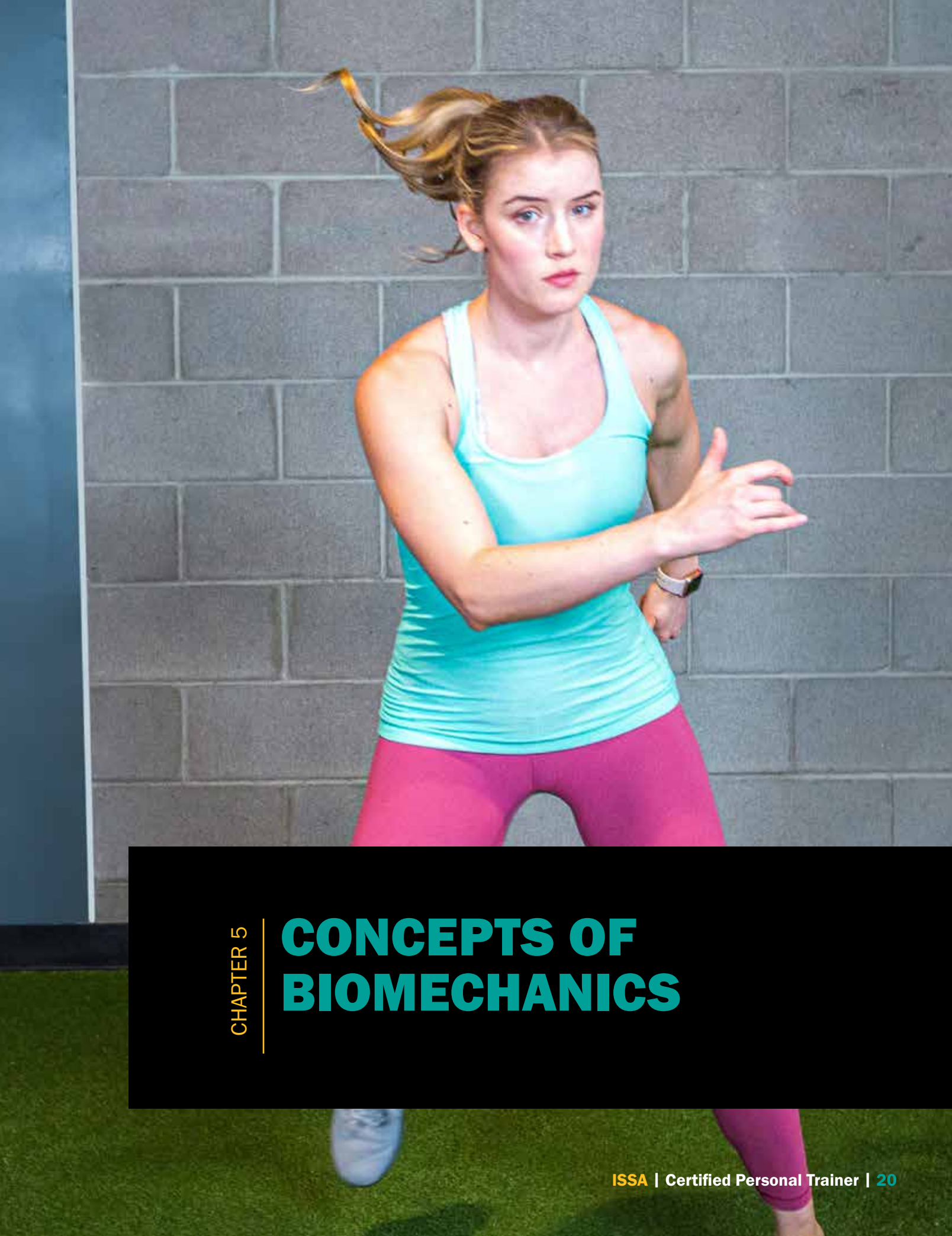
Name the structures of the respiratory system.

Define “internal respiration” and “external respiration.”

Explain the functions of cortisol in the body.

Describe the function of insulin in the body.

List the structures of the digestive system.



CHAPTER 5

CONCEPTS OF BIOMECHANICS

Fill in the blanks in the following table.

ANATOMICAL LOCATION TERM	DEFINITION
Anterior or ventral	
	Back of the body or toward the back relative to another reference point
Superior	
Inferior	
	Position closer to the center of the body relative to a reference point
Distal	Position farther from the reference point
Medial	
	Position relatively farther from the midline of the body
Prone	Lying facedown
	Lying on one's backside
Deep	
Superficial	Closer to the surface relative to another reference point
	Refers to only one side
	Refers to both sides
Ipsilateral	
	On the opposite side
Caudal	
	Toward the head
Volar	

Fill in the blanks in the following table.

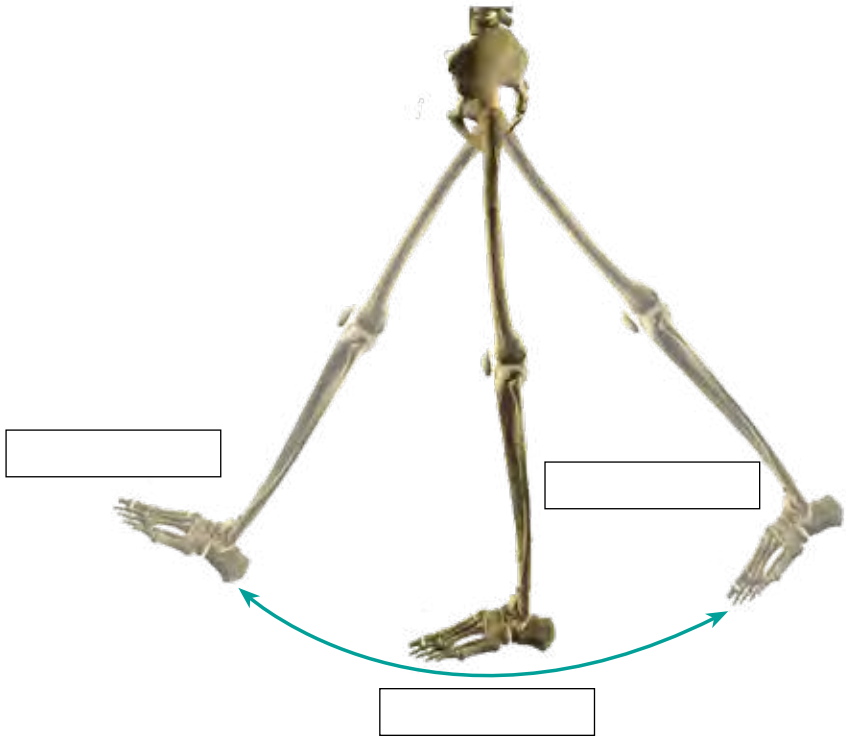
TERM	DEFINITION/ACTION
Abduction	
	Movement toward the midline
Flexion	Movement decreasing the angle between two body parts
Extension	
	Flexion in the frontal plane
	Abduction of the scapula
Retraction	
Elevation	Movement in a superior direction
Depression	
	Extension of the foot downward (inferiorly)
Dorsiflexion	Flexion of the foot upward (superiorly)
External rotation	
	Rotational movement toward the midline
Circumduction	Circular movement of a limb extending from the joint where the movement is controlled
Inversion	
Eversion	Movement of the sole of the foot away from the median plane
	Turning the palm or arch of the foot down
Supination	
Hyperextension	Position that extends beyond anatomical neutral
	Same-side movement
Contralateral	Opposite-side movement
Lateral	
	Situated toward or closer to the midline

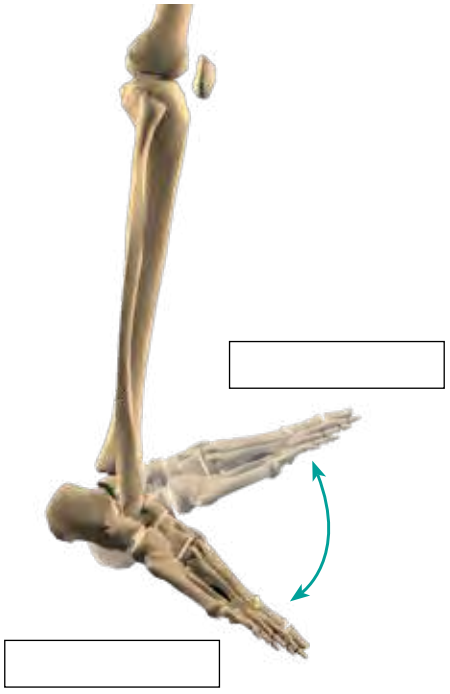
Define the three imaginary planes of motion and provide an exercise that is performed within each plane.

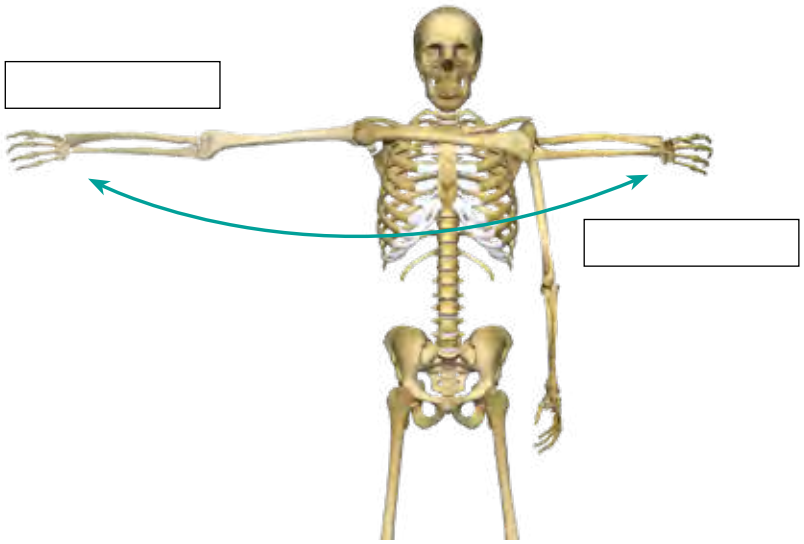
Explain the relationship between agonist and antagonist.

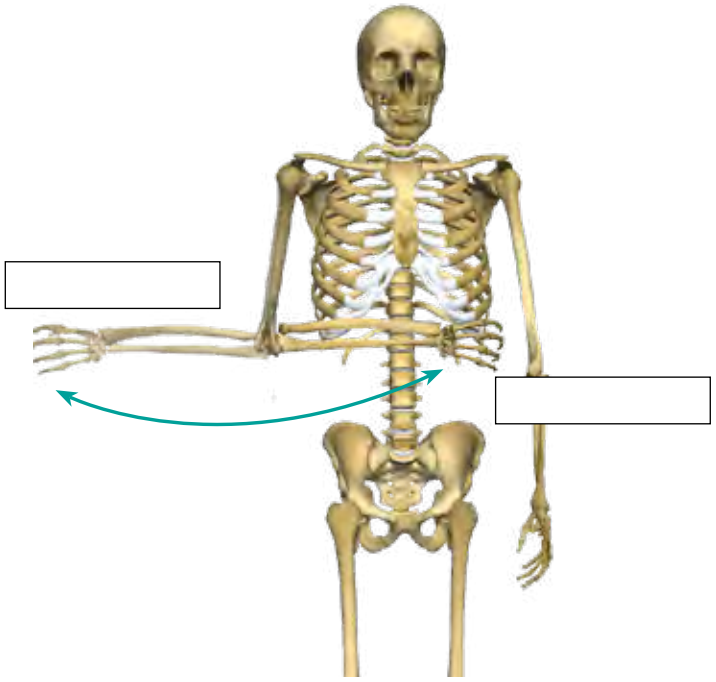
List and explain the three levers in the body and provide an example of each.

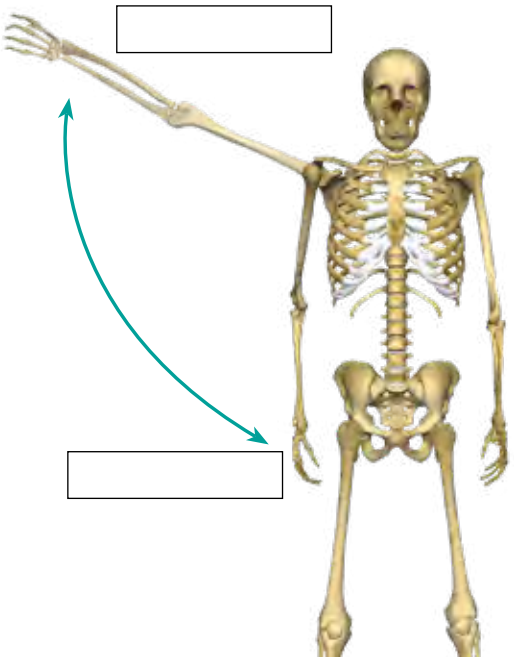
Label the following anatomical movements.











Define "balance."

Define "equilibrium."

Define "stability."

Name and explain Newton's First Law.

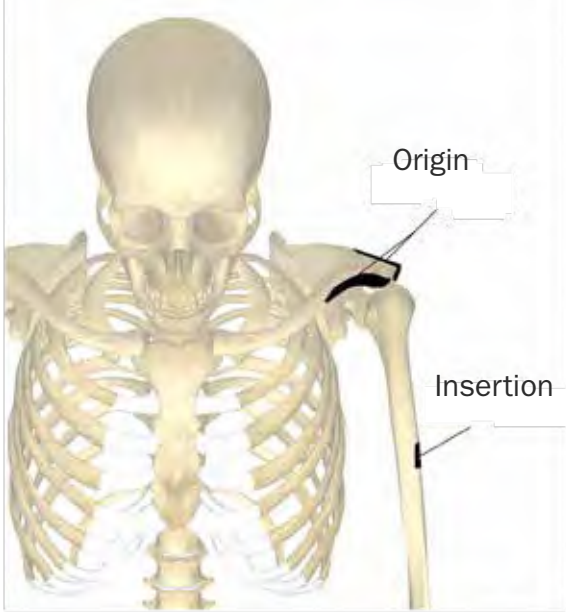
Name and explain Newton's Second Law.

Name and explain Newton's Third Law.

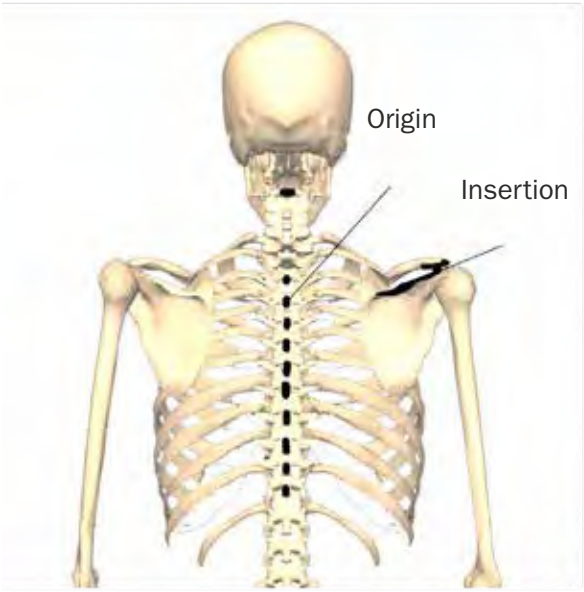
Define "center of gravity" and explain how it relates to movement.

Identify the muscles with the following insertions and origins.

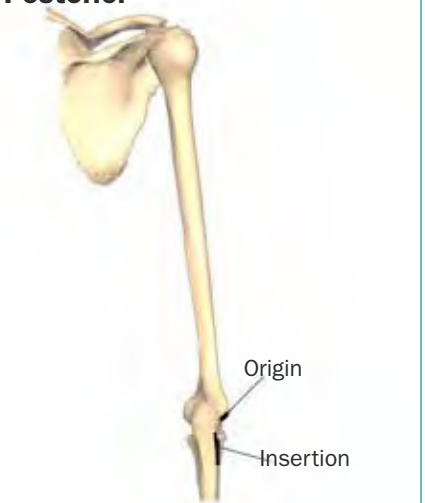
Anterior



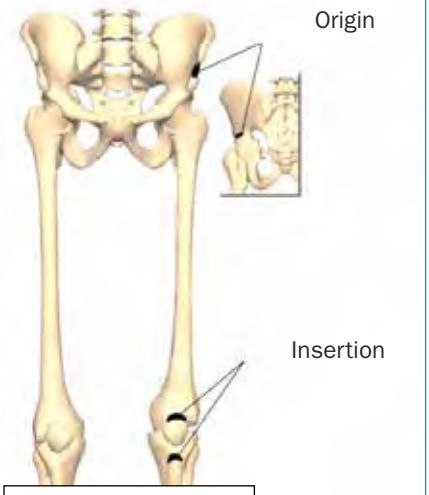
Posterior



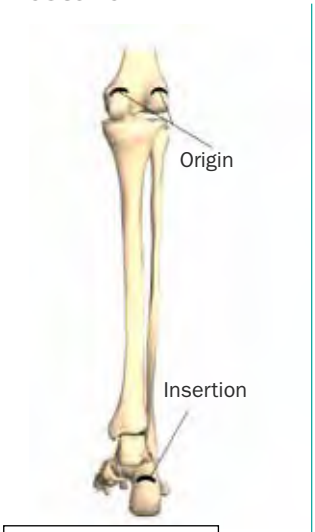
Posterior



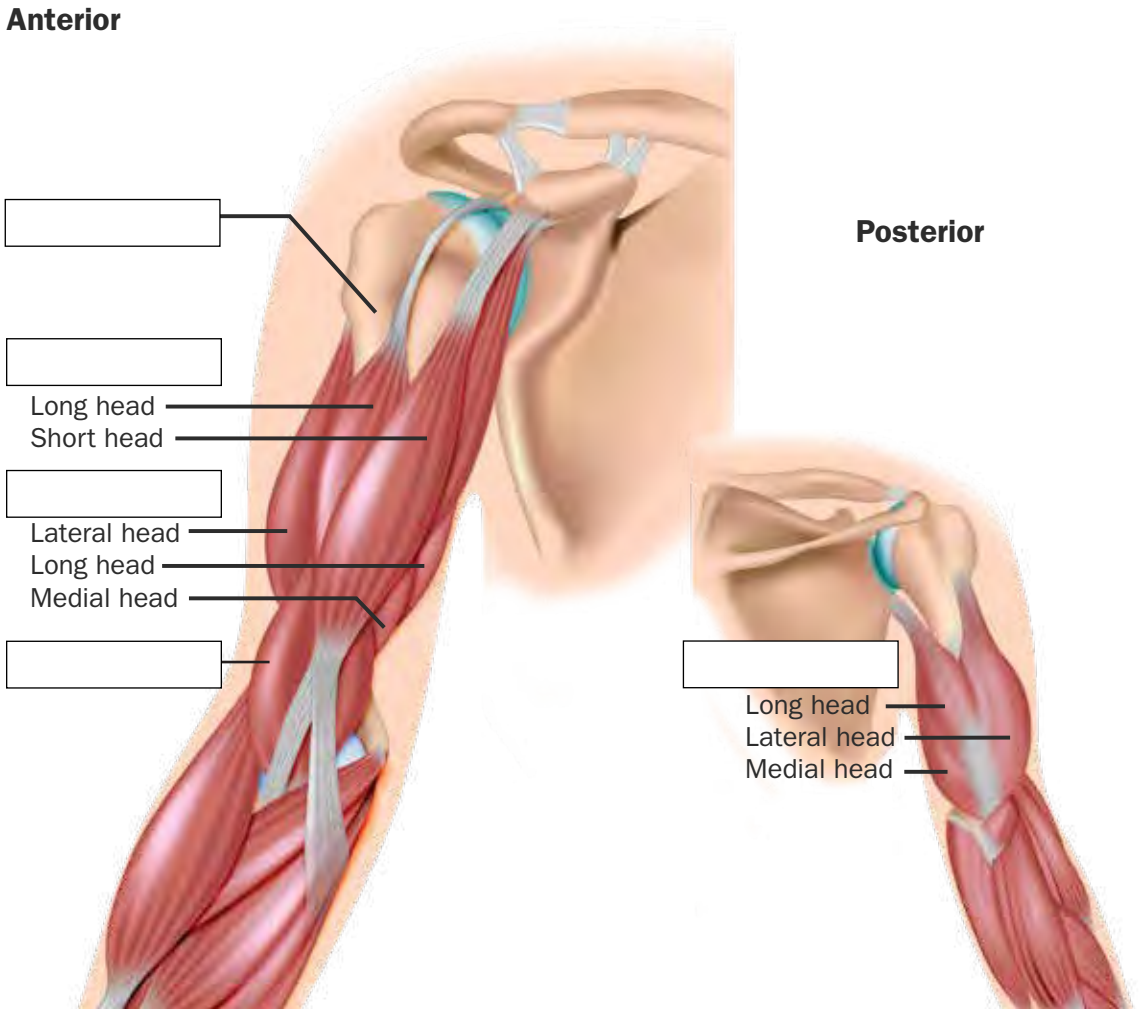
Anterior



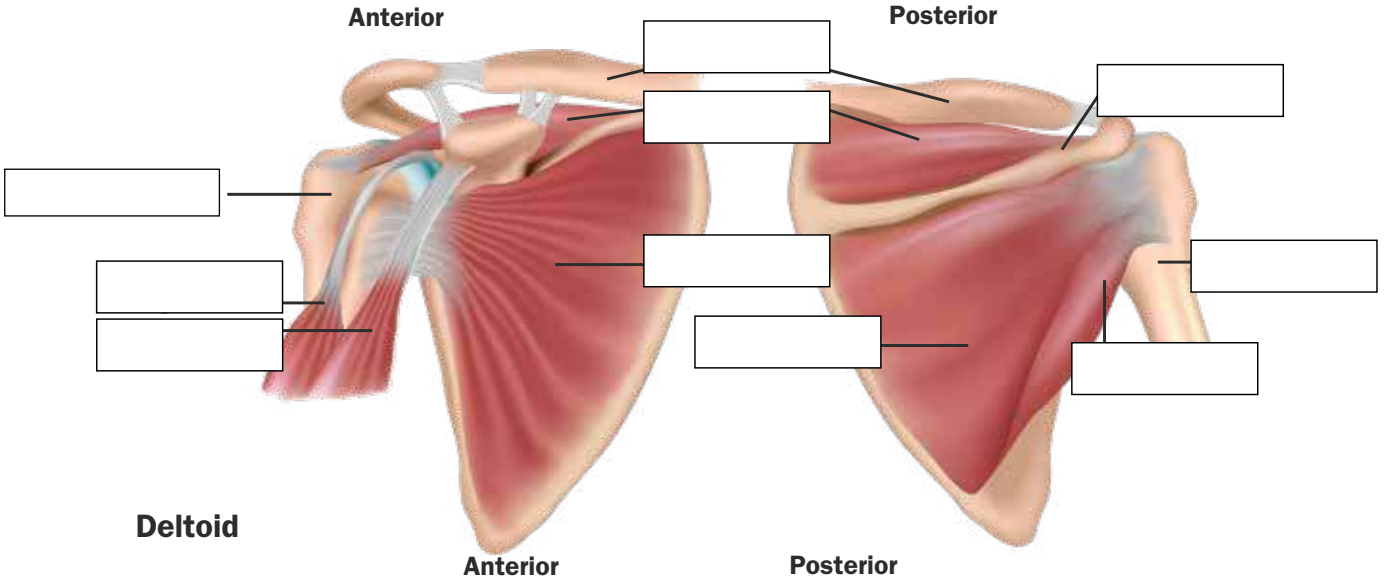
Posterior



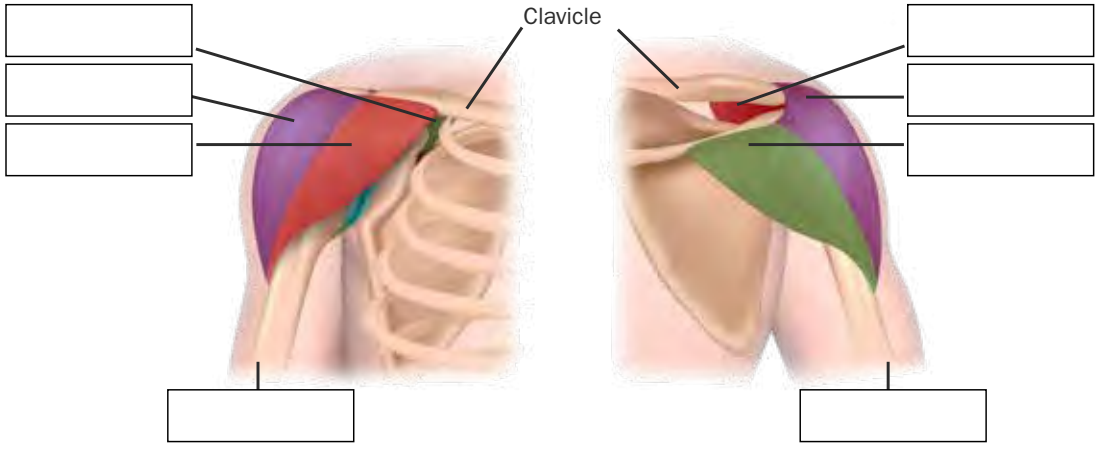
Label the following muscles by body region.

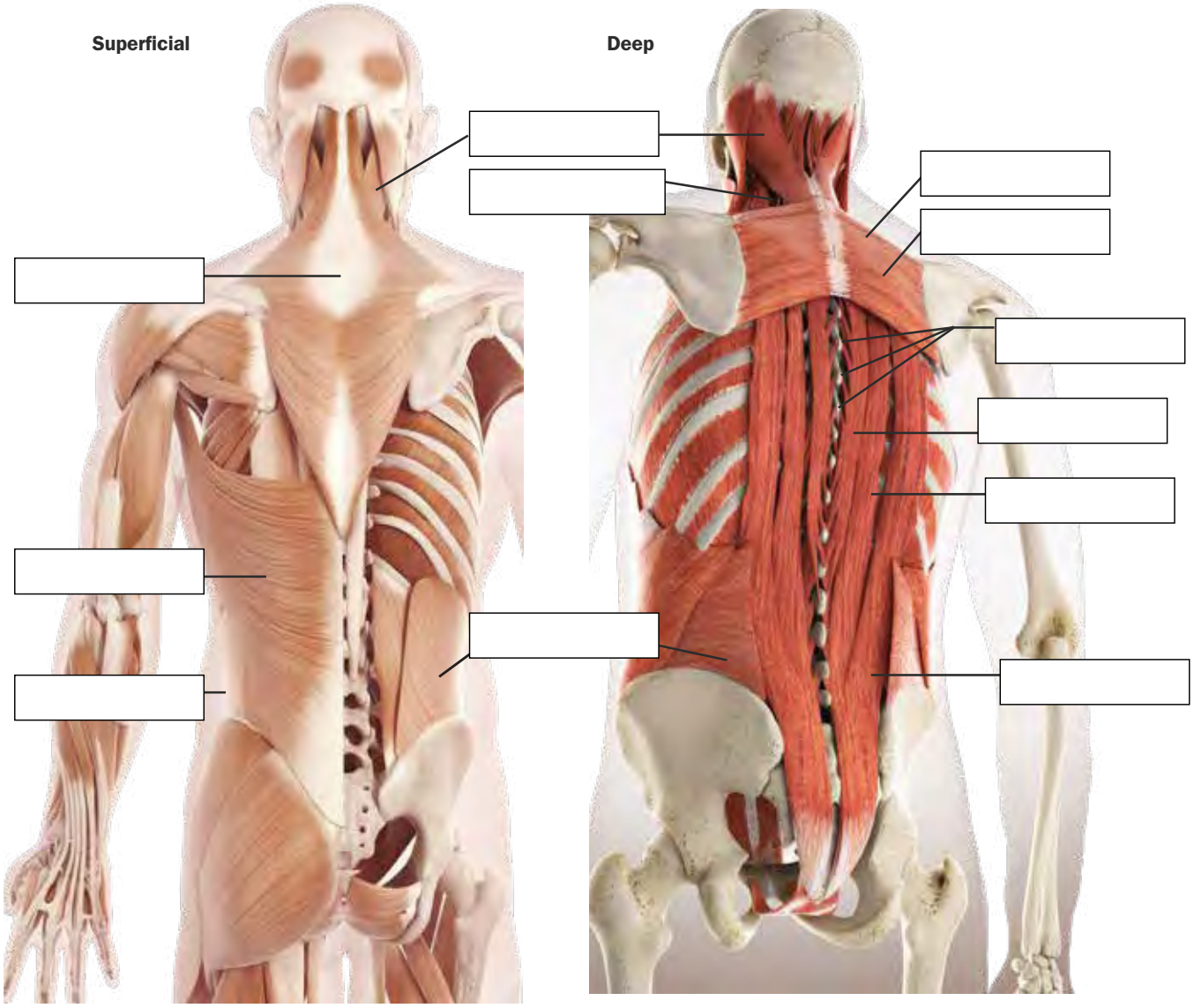


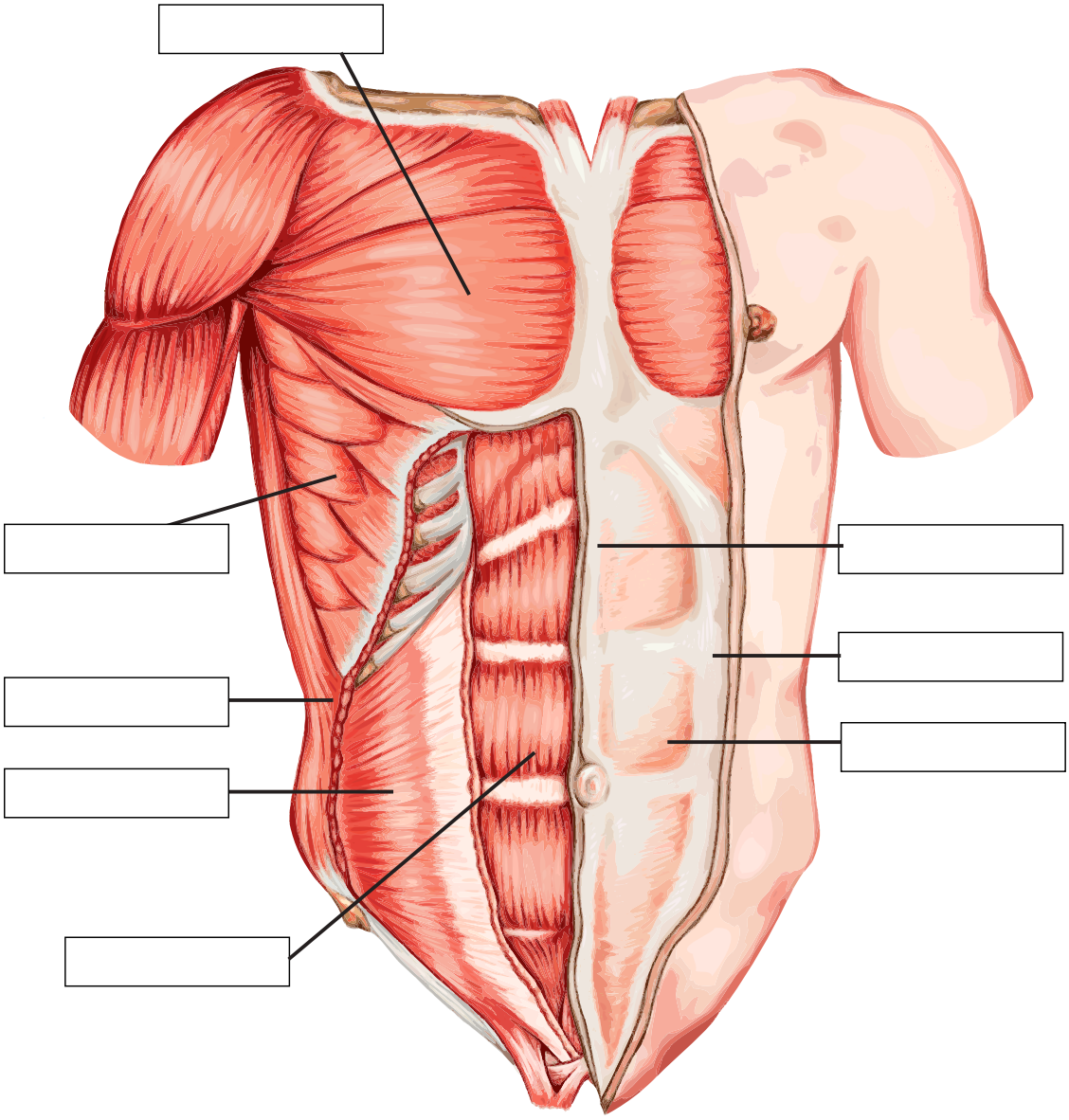
Shoulder

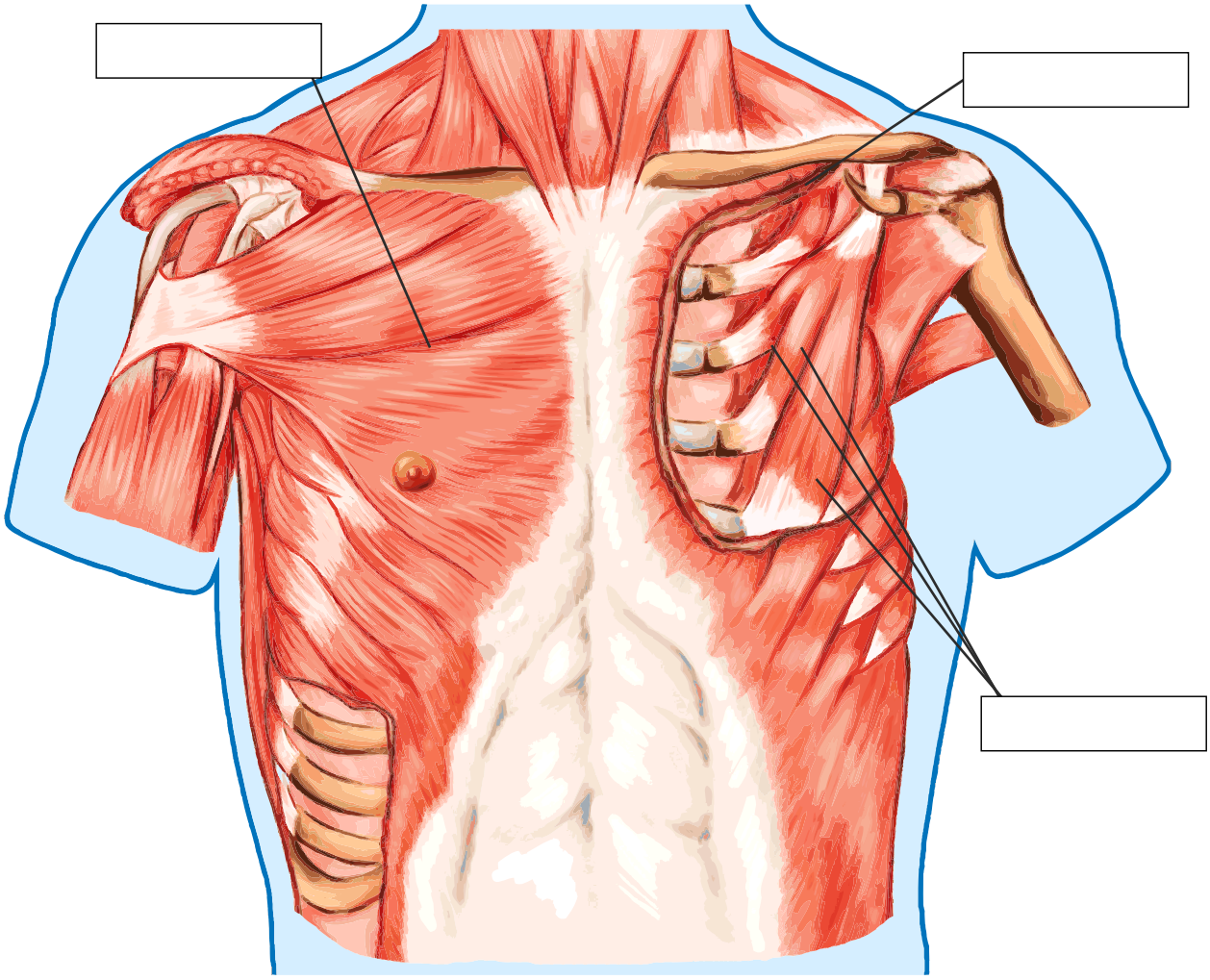


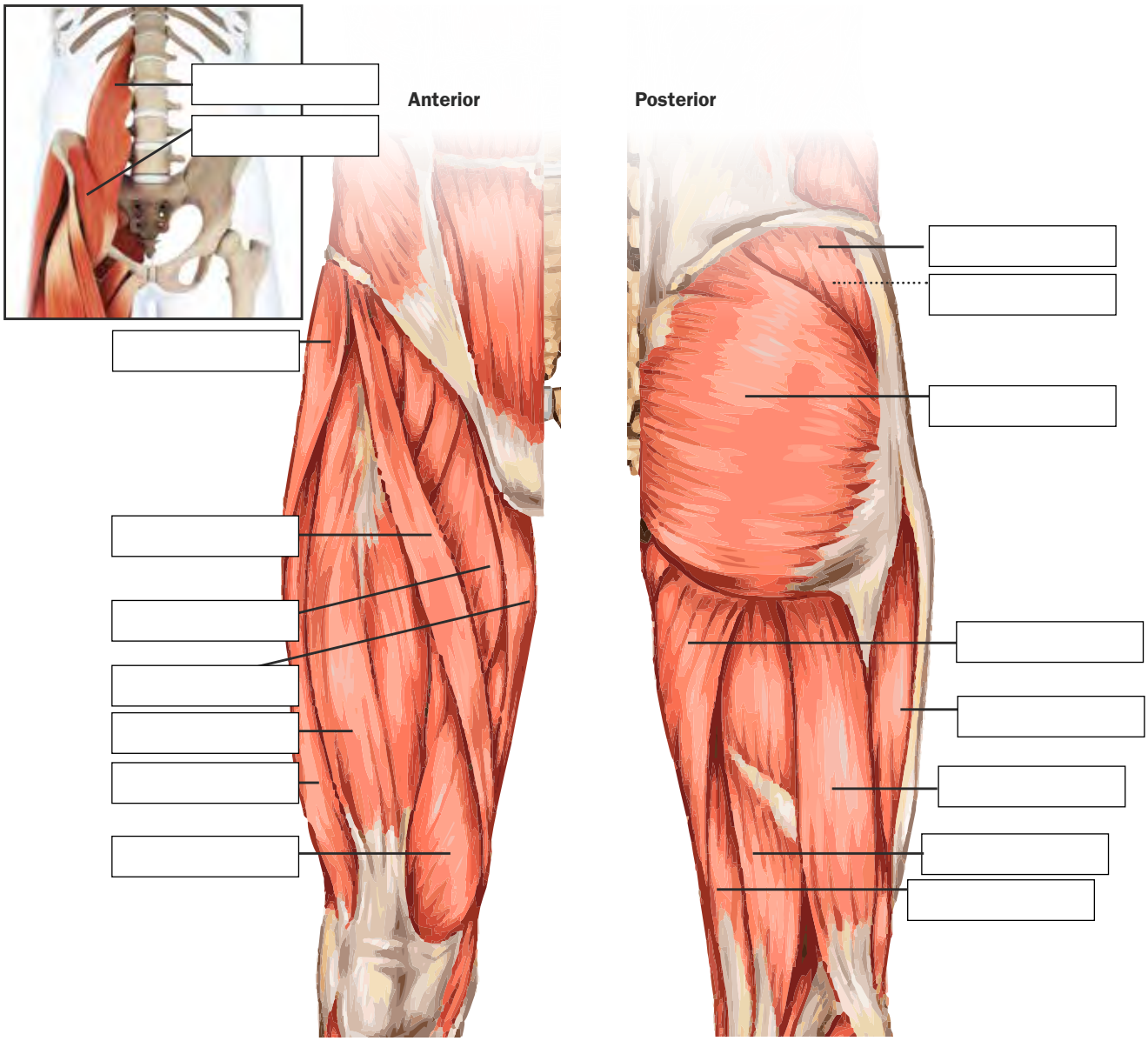
Deltoid

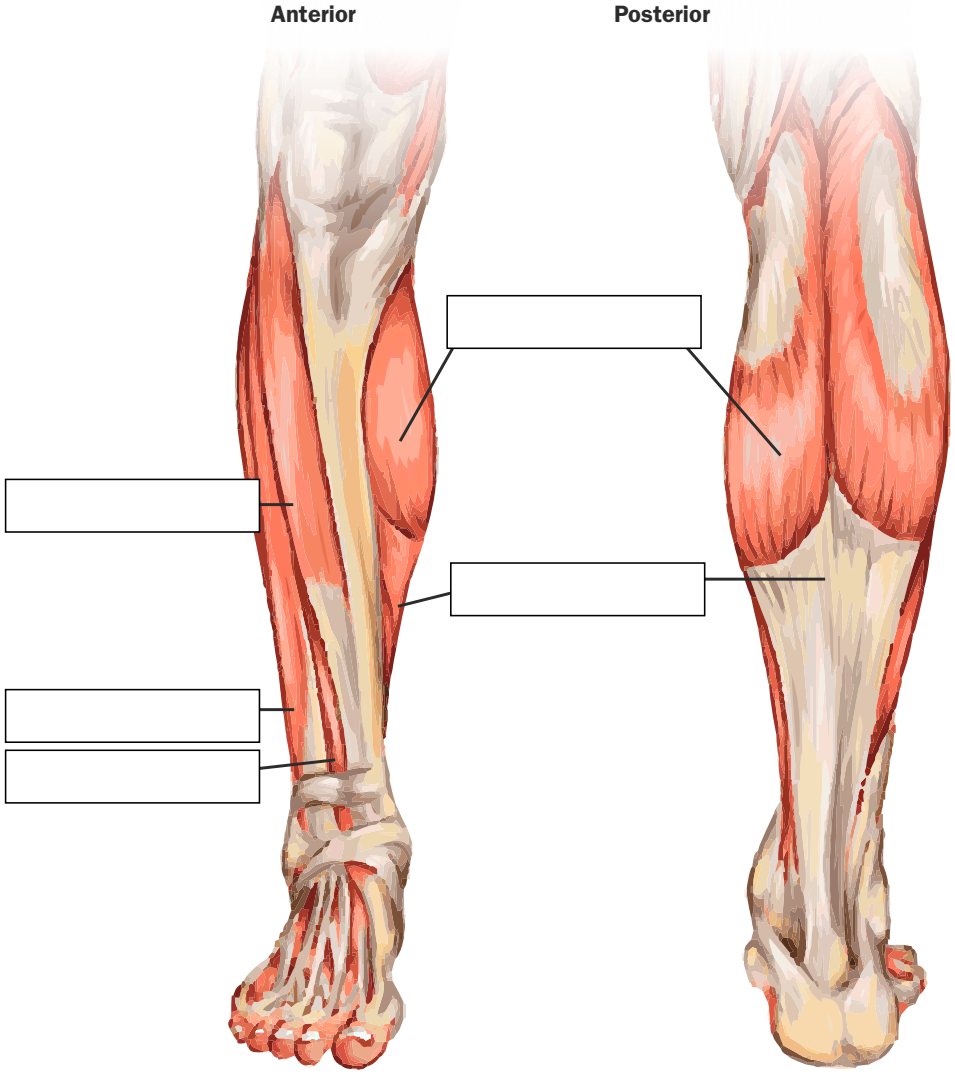



Superficial
Deep






Anterior
Posterior


Anterior
Posterior

WORKBOOK

List the muscles of the rotator cuff.

Which bones make up the shoulder girdle?

Which bones make up the elbow joint?

Which bones make up the knee joint?



CHAPTER 6

ENERGY AND METABOLISM

WORKBOOK

List the organelles that make up the cell.

Where does the energy come from that fuels our physical activity?

What is “homeostasis”?

WORKBOOK

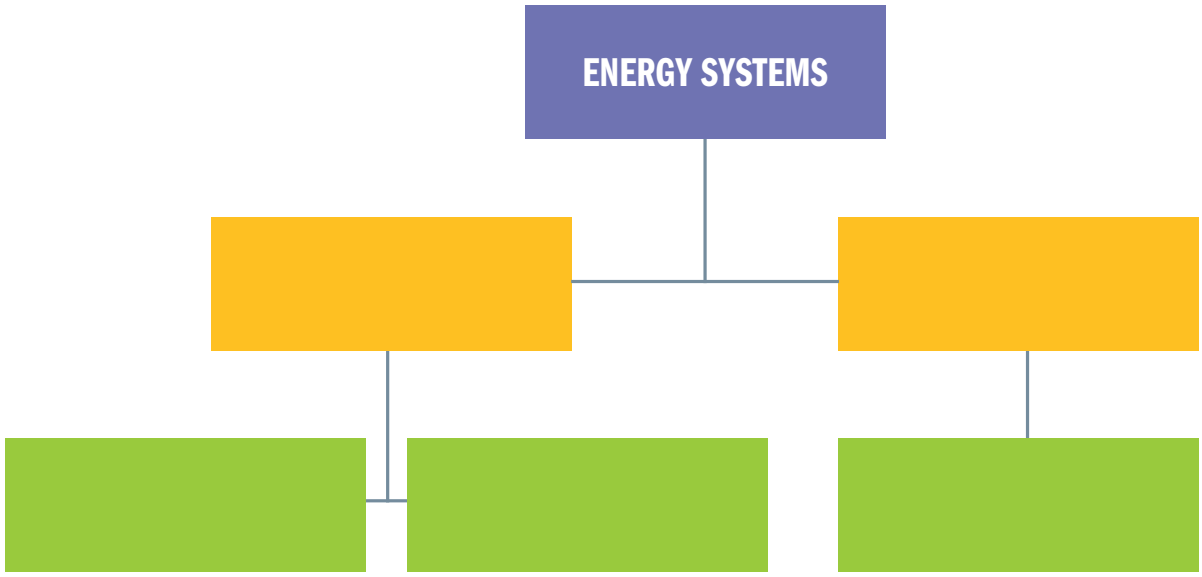
Define "metabolism."

List and define the two phases of metabolism.

What is RMR?

What is the thermic effect of food?

Fill in the following chart.



Fill in the following table.

SOURCE	CAL YIELD PER GRAM
Nutritional carbohydrate	
Nutritional protein	
Nutritional fat	

Use the following table to answer the question below.

CALCULATING CALORIC EXPENDITURE	
MALE	metric: **DCE=ALF x [(13.75 x WKG) + (5 x HC) - (6.76 x age) + 66]** imperial: **DCE=ALF x [(6.25 x WP) + (12.7 x HI) - (6.76 x age) + 66]**
FEMALE	metric: **DCE=ALF x [(9.56 x WKG) + (1.85 x HC) - (4.68 x age) + 655]** imperial: **DCE=ALF x [(4.35 x WP) + (4.7 x HI) - (4.68 x age) + 66]**

WHERE	
ALF = Activity level factor	**AND ALF HAS THE FOLLOWING VALUES**
DCE = Daily caloric expenditure	Sedentary: ALF = 1.2
HC = Height in centimeters	Lightly active: ALF = 1.375
HI = Height in inches	Moderately active: ALF = 1.55
WKG - Weight in kilograms	Very active: ALF = 1.725
WP = Weight in pounds	Extremely active: ALF = 1.9

Calculate the estimated caloric expenditure for a male client with the following information:

Sex- Male

Activity- Lightly active

Age- 33 years old

Weight-165 lbs (74.8 kg)

Height- 6'2" (74 inches or 188 cm)

Define each of the following terms and fill in the chart.

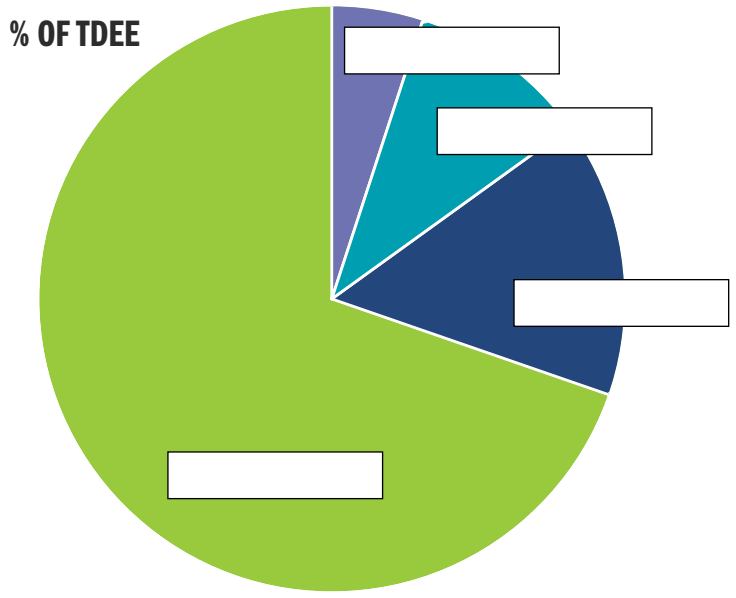
TDEE - _____

EAT - _____

TEF - _____

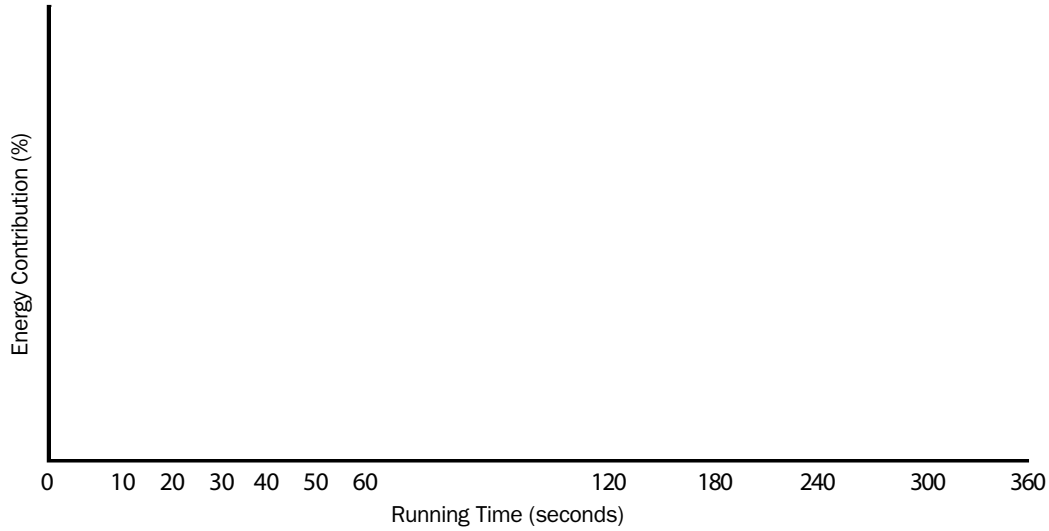
NEAT - _____

RMR - _____



Define “ATP” and briefly discuss the three energy pathways that produce ATP.

Fill in the graph below with the three pathways of energy production.



ATP/CP System
- 0-10 seconds
- 100m sprint

Glycolytic System
- 10-120 seconds
- 400m swim

Aerobic System
- Over 2 minutes
- Marathon



CHAPTER 7

CLIENT ASSESSMENTS

What information is collected on the following client forms?

Client Intake Form: _____

PAR-Q: _____

Health History Questionnaire: _____

3-Day Dietary Record: _____

How is BMI calculated?

_____

_____

_____

_____

WORKBOOK

Name the 7 sites for circumference or girth measurement.

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

Define "VO_2 max" and explain how it is used in fitness.

_____

_____

_____

_____

_____

_____

_____

List the kinetic chain checkpoints.

_____

_____

_____

_____

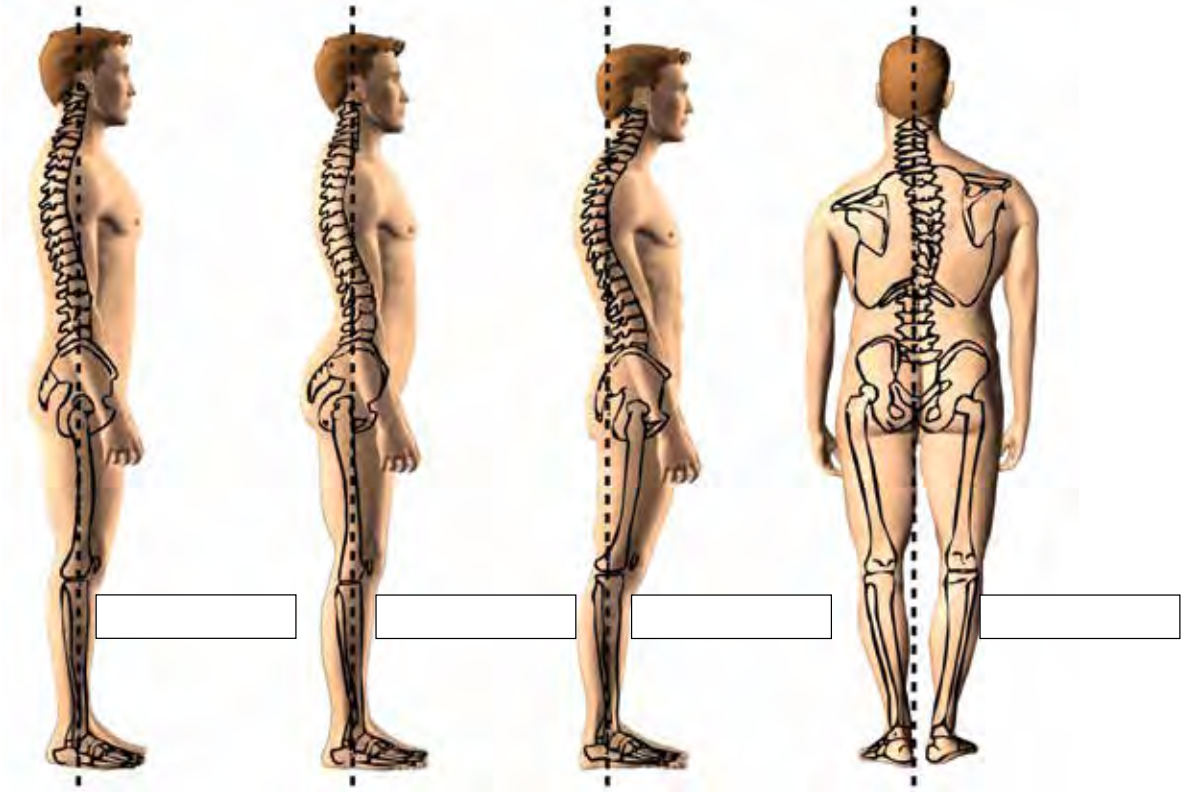
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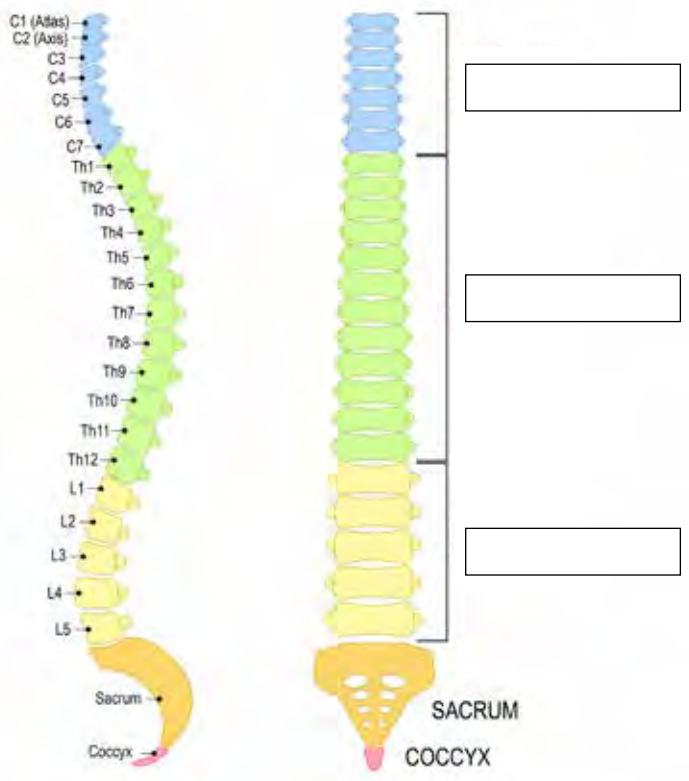
Identify the following postural deviations.



Fill in the following table.

MALALIGNMENT	POSSIBLE TIGHT MUSCLES	POSSIBLE WEAK MUSCLES
Lordosis		
Flat back		
Swayback		
Kyphosis		
Forward Head		

Identify the 3 main sections of the spine.



Fill in the table below for the potential observations from a squat assessment.

MALALIGNMENT	POSSIBLE WEAK MUSCLES	POSSIBLE TIGHT MUSCLES
	Gluteus maximus Gluteus medius Vastus medialis	Adductors Biceps femoris TFL Vastus lateralis
Foot turnout (eversion and external rotation)	Gastrocnemius (medial) Semitendinosus Semimembranosus Sartorius Gracilis	
	Gluteus maximus Erector spinae Anterior tibialis	Soleus Gastrocnemius Psoas Rectus femoris Rectus abdominis External obliques
Lumbar extension	Gluteus maximus Transverse abdominus Hamstring complex Multifidus	Psoas Rectus femoris TFL Erector spinae Latissimus dorsi
Heel elevation	Tibialis (anterior)	
Foot pronation (arch collapse)		Gastrocnemius (lateral) Biceps femoris TFL
	Gluteus medius (same side of shift) Tibialis (anterior) Adductors (opposite side of shift)	Adductors TFL (same side of shift) Gastrocnemius Soleus Piriformis Gluteus medius Biceps femoris (opposite side of shift)



CHAPTER 8

ELEMENTS OF FITNESS

WORKBOOK

Define "cardiovascular endurance."

Define "muscular strength."

Define "muscular endurance."

Define "flexibility."

Define "body composition."

WORKBOOK

List and explain the 8 elements of a fitness program.

1. _____.

_____

2. _____

_____

3. _____

_____

4. _____

_____

5. _____

_____

6. _____

_____

7. _____

_____

8. _____

_____

What formula is used to estimate maximum heart rate?

_____

_____

_____

_____

What three measures are used to determine the intensity of cardiorespiratory training?

_____

_____

_____

_____



CHAPTER 9

PRINCIPLES OF PROGRAM DESIGN

Define "acute training variables."

_____

_____

_____

_____

Explain the following acute training variables:

Type: _____

Exercise selection: _____

Exercise order: _____

Intensity: _____

Sets: _____

Repetitions: _____

Frequency: _____

Range of motion: _____

Time: _____

Tempo: _____

Time under tension: _____

Rest: _____

Identify the appropriate number of sets by training goal.

TRAINING GOAL	SETS
Muscular endurance	
Hypertrophy	
Maximum strength	
Power	

Identify the appropriate number of repetitions by training goal.

TRAINING GOAL	REPETITIONS
Muscular endurance	
Hypertrophy	
Maximum strength	
Power	

Identify the appropriate rest between sets by training goal.

TRAINING GOAL	REST BETWEEN SETS
Muscular endurance	
Hypertrophy	
Maximum strength	
Power	

Define the "Principle of Specificity."

_____

_____

_____

WORKBOOK

Define the "SAID Principle."

Define the "Principle of Individual Differences."

Define the "Principle of Progressive Overload."

Name the stages of the GAS Principle and give a brief explanation of each.

Name the three periods of training periodization and the length of each.

_____

_____

_____

_____

Fill in the following training progressions.

- [ ] ▶ unstable
- [ ] ▶ dynamic
- Unloaded ▶ [ ]
- [ ] ▶ fast
- Simple ▶ [ ]

Define "overreaching" and "overtraining."

_____

_____

_____

_____

Identify the elements of fitness associated with each training category.

TRAINING CATEGORY	FITNESS PROGRAM ELEMENT(S) AND TYPES
Foundational training	
Strength training	
Metabolic training	
Speed, agility, and quickness training	



CHAPTER 10

CONCEPTS OF FLEXIBILITY

WORKBOOK

Name 4 benefits of flexibility training.

1. _____

2. _____

3. _____

4. _____

Define "passive stretching."

_____

_____

_____

Define "dynamic stretching."

_____

_____

_____

Define "active stretching."

_____

_____

_____

WORKBOOK

List 5 dynamic stretches.

1. _____
2. _____
3. _____
4. _____
5. _____

List 5 static stretches.

1. _____
2. _____
3. _____
4. _____
5. _____

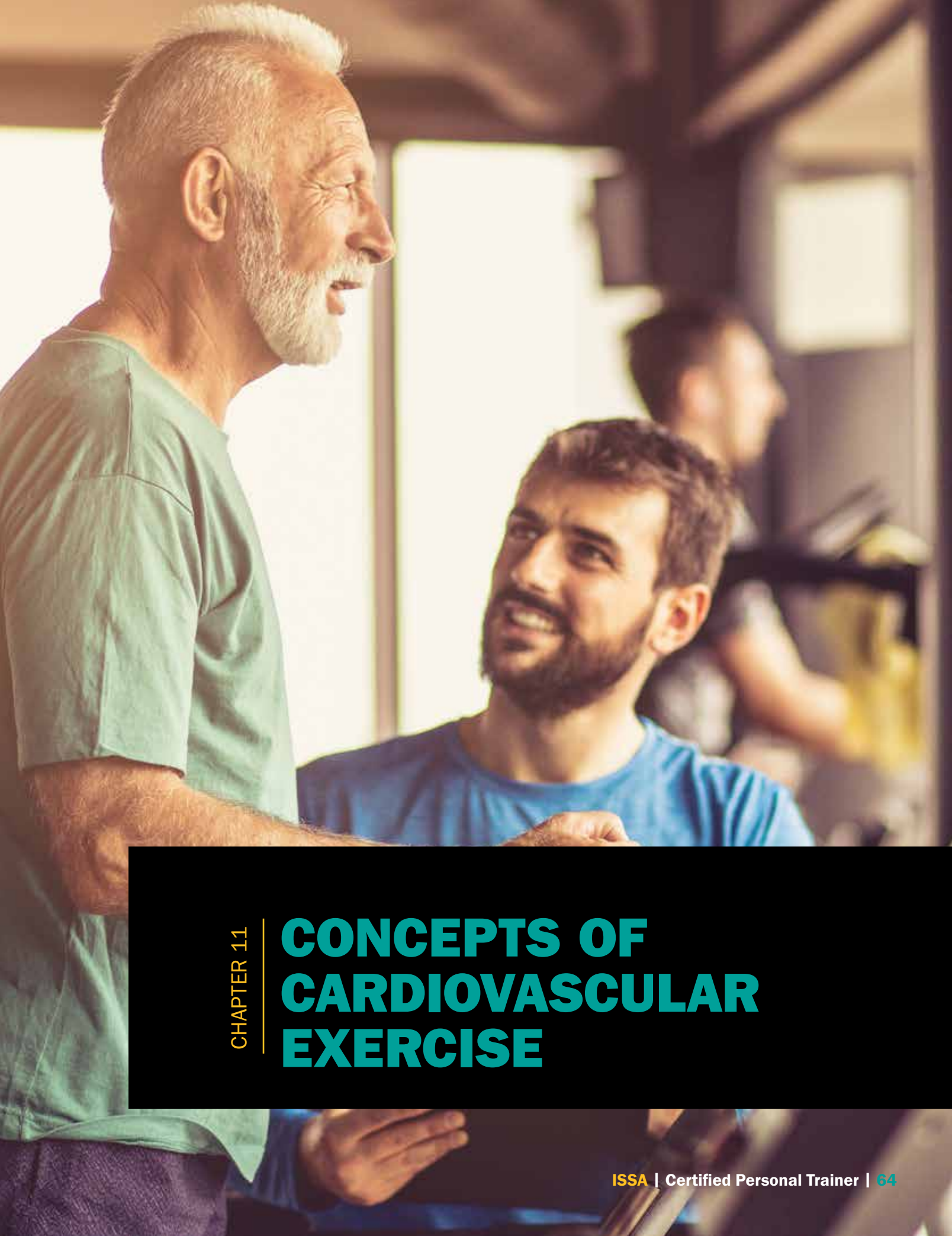
Define “myofascial release.”

_____

_____

_____

_____



CHAPTER 11

CONCEPTS OF CARDIOVASCULAR EXERCISE

List 6 benefits of cardiovascular exercise.

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

What does the F.I.T.T. acronym stand for as it relates to cardiovascular training?

F - _____

I - _____

T - _____

T - _____

Define "vital capacity."

_____

_____

_____

_____

Define "tidal volume."

Define "metabolic equivalent (MET)."

List the formula used to calculate metabolic equivalent (MET).

Fill in the appropriate percentages of max heart rate that correspond with the heart rate zones below.

ZONE 1 · _____ Max HR
Very light activity, such as warm-up/cooldown

ZONE 2 · _____ Max HR
Light activity, such as slow-paced jogging, walking up a flight of stairs, lightweight low resistance

ZONE 3 · _____ Max HR
Moderate activity that increases aerobic endurance, such as moderate jogging, cycling, or rowing

ZONE 4 · _____ Max HR
Hard anaerobic activity, such as high rep ball slams, boxing, or heavy weight lifting

ZONE 5 · _____ Max HR
Extreme hard maximum exertion activity, such as sprinting. All out effort!

Explain the physiological adaptations that can occur with cardiorespiratory training.

Explain "circuit training."

Fill in the following energy systems table.

ENERGY SYSTEM	WHEN SYSTEM IS DOMINANT	TYPICAL ACTIVITIES	TIME REQUIRED FOR FULL RECOVERY
	0-10 sec.	Resistance training <200 M sprints Plyometrics and ballistics	
Glycolytic		Badminton Soccer Gymnastics Hockey 100–400 M sprints	20–60 min.
Aerobic	2 min. and longer		



CHAPTER 12

CONCEPTS OF RESISTANCE TRAINING

Fill in the following table of categories of strength.

STRENGTH CATEGORY	DEFINITION
	The ability to recruit as many motor units as possible instantaneously at the start of a movement
	Determined by considering the individual's body weight in relation to the amount of resistance they can overcome and found with the following calculation: 1RM / body weight = force per unit of body weight
Maximum strength	
	The combination of strength and speed—the ability for a muscle to generate maximal tension as quickly as possible
Speed strength	

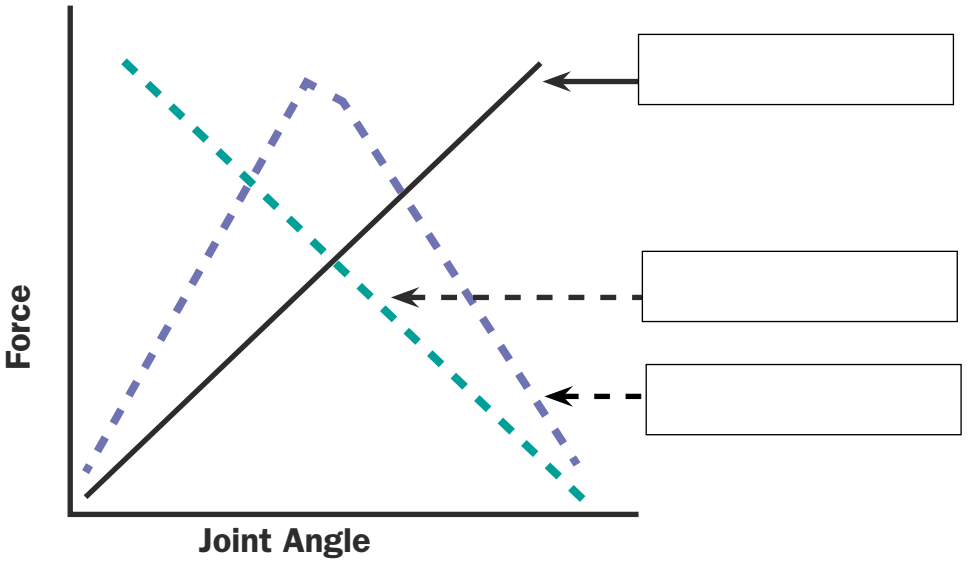
Name 3 benefits of resistance training.

1. _____

2. _____

3. _____

Label the following strength curve.



Define “isolation exercises” and give an example.

_____

_____

_____

Define “compound exercises” and give an example.

_____

_____

_____

Name the three phases of an explosive movement.

Phase 1 - _____

Phase 2 - _____

Phase 3 - _____

Name 3 benefits and 3 drawbacks of free weights.

	Benefits	Drawbacks
1.	_____	_____
2.	_____	_____
3.	_____	_____

Name 3 benefits and 3 drawbacks of weight machines.

	Benefits	Drawbacks
1.	_____	_____
2.	_____	_____
3.	_____	_____

Define a "superset."

_____

_____

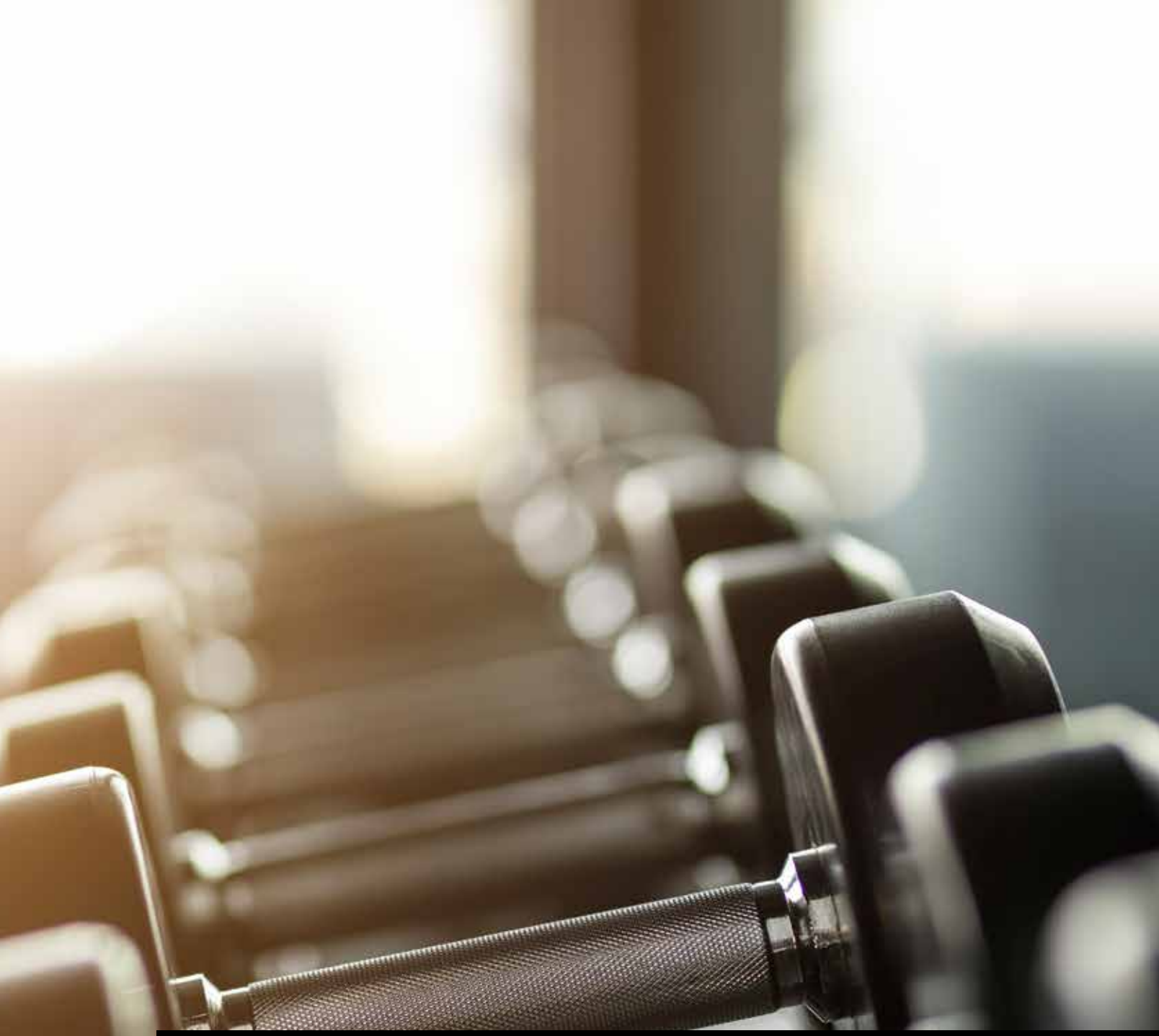
_____

Define a "drop set."

_____

_____

_____



CHAPTER 13

EXERCISE SELECTION AND TECHNIQUE

Explain “exercise progression” and “exercise regression.”

Name the three different learning styles.

1. _____
2. _____
3. _____

Explain how each learning style best learns information.

WORKBOOK

Name the 6 movement categories and give an example of an exercise for each category.

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____



What exercise is shown in the images above?

_____

What is/are the prime mover(s) for this exercise?

_____

Name an alternative exercise for the prime mover(s) involved in this exercise.

_____



What exercise is shown in the images above?

_____

What is/are the prime mover(s) for this exercise?

_____

Name an alternative exercise for the prime mover(s) involved in this exercise.

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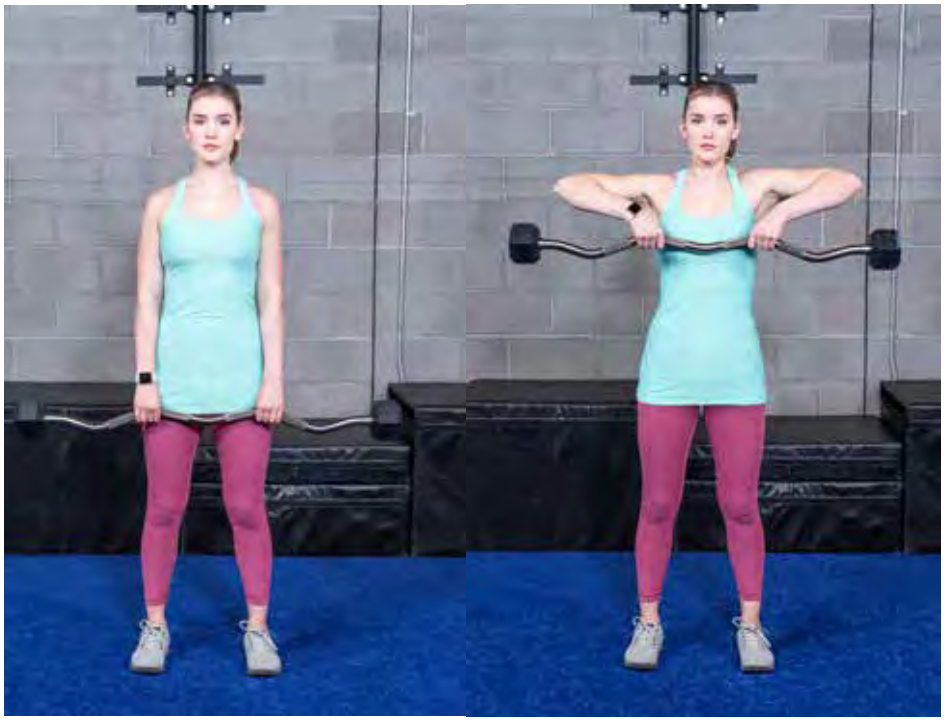
_____

What is/are the prime mover(s) for this exercise?

_____

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_____



What exercise is shown in the images above?

_____

What is/are the prime mover(s) for this exercise?

_____

Name an alternative exercise for the prime mover(s) involved in this exercise.

_____



What exercise is shown in the images above?

_____

What is/are the prime mover(s) for this exercise?

_____

Name an alternative exercise for the prime mover(s) involved in this exercise.

_____



CHAPTER 15

NUTRITION FOUNDATIONS

WORKBOOK

Define "macronutrient" and "micronutrient."

_____

_____

_____

Name the three macronutrients.

1. _____
2. _____
3. _____

Which macronutrient is the primary source of energy for the body?

_____

Explain what each macronutrient is broken down to in the body.

_____

_____

_____

_____

_____

Name an example of each:

A simple carbohydrate: _____

A complex carbohydrate: _____

Give the general intake recommendations for each macronutrient (as a percentage of total daily calories or amount per body weight).

_____

_____

_____

_____

_____

_____

Name the 9 essential amino acids.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____

WORKBOOK

What percentage of water in the body is intracellular and what percentage is extracellular?

_____

_____

_____

What is the general recommendation for water intake daily for an adult?

_____

Name 3 macrominerals.

1. _____
2. _____
3. _____

Describe the ketogenic diet.

_____

_____

_____

What foods are avoided in a gluten-free diet?

_____

_____

_____

Name 5 nutritional limiting factors that may keep a client from seeing fitness results.

1. _____

2. _____

3. _____

4. _____

5. _____

What are the portion size recommendations for the following types of foods:

Men:

Protein dense food: _____

Vegetables: _____

Carbohydrates: _____

Fat-dense food: _____

Women:

Protein dense food: _____

Vegetables: _____

Carbohydrates: _____

Fat-dense food: _____



CHAPTER 16

SUPPLEMENTATION

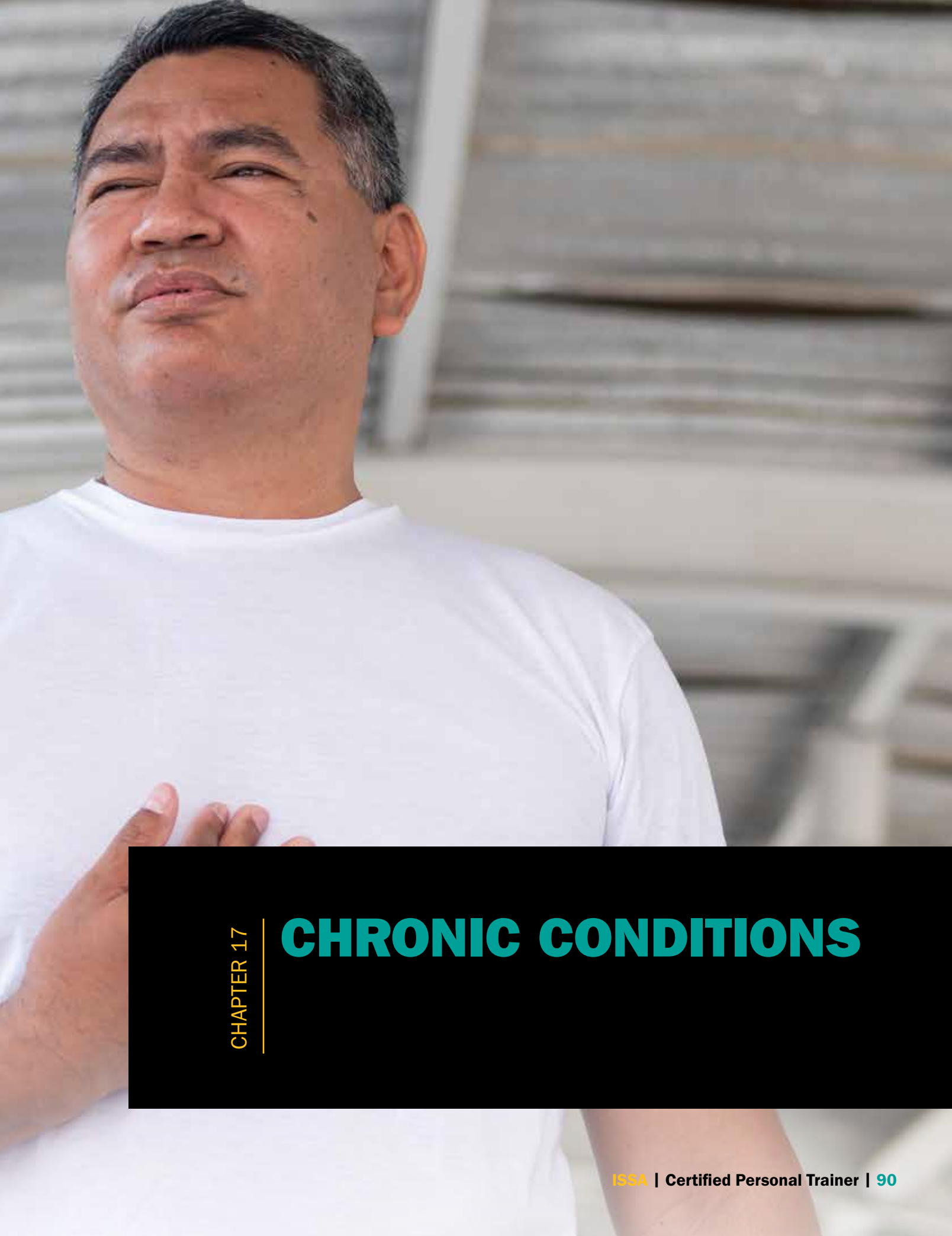
WORKBOOK

Define a "dietary supplement."

Define "ergogenic aid."

What is the most common nutritional supplement?

What is the scope of practice for a personal trainer regarding supplementation?



CHAPTER 17

CHRONIC CONDITIONS

WORKBOOK

Define "hypertension."

_____

_____

_____

Name 3 risk factors for hypertension.

1. _____
2. _____
3. _____

Define "type 2 diabetes."

_____

_____

_____

Define "type 1 diabetes."

_____

_____

_____

Name 3 risk factors for type 2 diabetes.

1. _____
2. _____
3. _____.

Define "arthritis."

_____

_____

_____

Name 3 risk factors for arthritis and related conditions.

1. _____
2. _____
3. _____

Describe program design considerations for a client with arthritis.

_____

_____

_____

_____

_____

Name 3 risk factors for coronary heart disease (CHD).

1. _____
2. _____
3. _____

Name 3 common triggers for asthma.

1. _____

2. _____

3. _____

Name 3 risk factors for asthma.

1. _____

2. _____

3. _____

Describe program design considerations for a client with asthma.

_____

_____

_____

_____

_____

_____



CHAPTER 18

LIFESPAN POPULATIONS

Name 3 ways to prevent injury in youth clients.

1. _____

2. _____

3. _____

Name 4 differences between youth and adults.

1. _____

2. _____

3. _____

4. _____

Name 2 aerobic (cardiovascular) assessments for youth clients.

1. _____

2. _____

Name 3 client assessments that can be used with senior clients.

1. _____

2. _____

3. _____

List 3 absolute contraindications for a pregnant client.

1. _____

2. _____

3. _____

List 3 relative contraindications for a pregnant client.

1. _____

2. _____

3. _____



CHAPTER 19

BUSINESS AND MARKETING

WORKBOOK

Name and explain the 6 styles of personal training.

1. _____

_____

2. _____

_____

3. _____

_____

4. _____

_____

5. _____

_____

6. _____

_____

Name 2 benefits of buddy training.

_____

_____

_____

_____

WORKBOOK

Name and describe the 5 types of emails used in fitness marketing.

1. _____

_____

2. _____

_____

3. _____

_____

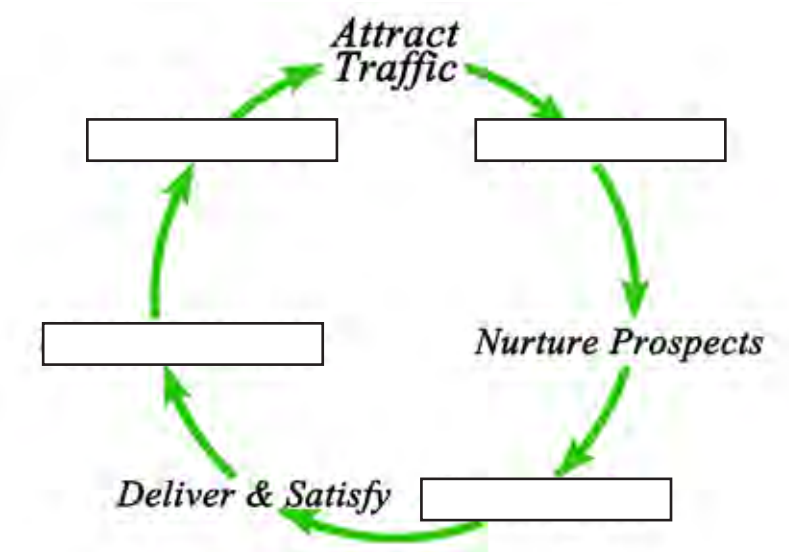
4. _____

_____

5. _____

_____

Fill in the aspects of the client lifecycle.



WORKBOOK

List the situations where a personal trainer should refer a client to an appropriate professional.

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

Explain the following 5 categories in a code of ethics.

1. Integrity: _____

_____

2. Objectivity: _____

_____

3. Professional competence: _____

_____

4. Confidentiality: _____

_____

5. Professional behavior: _____

_____



CHAPTER 20

SAFETY AND EMERGENCY SITUATIONS

Name the three responsibilities of a fitness professional related to cleaning duties.

1. _____

2. _____

3. _____

Define the following terms.

Cleaning: _____

_____

_____

Disinfecting: _____

_____

_____

Sanitizing: _____

_____

_____

Describe how gym equipment should be stored.

_____

_____

_____

_____

Name 6 things that can be found in a first aid kit.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

Describe the information that must be collected for an incident report.

_____

_____

_____

What does the acronym RICE stand for?

R - _____

I - _____

C - _____

E - _____

Name 5 emergency situations where 911 should be called immediately.

1. _____

_____

2. _____

_____

3. _____

_____

4. _____

_____

5. _____

_____